

Dyadic Frequency Laws, Clock Dynamics, and Defect Scaling in a Perturbed Hofstadter Q -Recursion

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Abstract

We study the perturbed Hofstadter Q -recursion defined by $Q(1) = Q(2) = 1$ and

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n \quad (n \geq 3).$$

We investigate the frequency structure and dyadic fluctuation geometry of the sequence. We prove an explicit dyadic law for the value frequencies:

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

The proof uses Cloître's binary interleaving structure, dyadic hitting-time identities, and an induced rank-lifting argument for plateau zero-runs.

To study deviations from linear growth, we introduce the renormalized defect

$$R(n) = Q(2n) - 2Q(n),$$

and prove the exact identity

$$R(n) = 2t_1(n + 1) - t_1(2n + 1) - 1,$$

which expresses the dyadic fluctuations through a single auxiliary clock process.

Extensive computations indicate structured fluctuations across dyadic scales together with possible logarithmic-scale organization on the $\log_2 n$ -scale. Combined with recent asymptotic results of Cloître showing that

$$Q(n) = \frac{n}{2} + O\left(\frac{n}{\sqrt{\log n}}\right),$$

these results suggest the presence of nontrivial recursive dyadic scaling behavior in the perturbed recursion.

Keywords. Nested recurrences; Hofstadter Q -sequence; meta-Fibonacci sequences; dyadic frequency laws; binary interleaving; plateau local times; clock dynamics; renormalized defects; logarithmic-scale fluctuations.

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1 Introduction

Nested recursions of Hofstadter type are among the most intricate and least understood objects in experimental mathematics. Their defining feature is self-reference: the value of the sequence at index n depends recursively on previous values of the sequence itself through nonlinear index shifts. Such recursions often generate behavior that appears irregular, chaotic, or only partially structured, and even elementary questions concerning existence, monotonicity, or asymptotic growth may remain unresolved for long periods of time.

The classical example is Hofstadter's Q -sequence $Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2))$, introduced in [5]. Despite its simple definition, the long-term behavior of the sequence is still not fully understood. Hofstadter-type recursions have since become a central source of examples in the theory of nested recurrences and meta-Fibonacci sequences. Related systems include the Conolly sequence [3], Tanny's recursion [12], and numerous variants connected to combinatorial trees, coding theory, and recursive structures [6, 4, 10, 9]. Many of these sequences exhibit a remarkable mixture of apparent disorder and hidden regularity.

One of the most striking phenomena in this area is the emergence of stable asymptotic laws from highly nonlinear recursive definitions. Several nested recursions are now known to possess linear or quasi-linear growth together with explicit frequency distributions [7, 1]. Nevertheless, the mechanisms generating these structures are often poorly understood, and explicit formulas remain comparatively rare.

In this paper we study the perturbed Hofstadter recursion defined by $Q(1) = Q(2) = 1$ and

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n, \quad n \geq 3,$$

introduced by the author and listed as sequence A394051 in the OEIS [8]. The perturbation by the alternating term $(-1)^n$ produces behavior that differs substantially from the classical Hofstadter sequence while still retaining its nested self-referential character.

Initial numerical experiments revealed several unexpected features of the perturbed recursion. The sequence appears to satisfy asymptotically linear growth with slope $1/2$, while simultaneously exhibiting structured dyadic fluctuations. In addition, the value frequencies admit explicit closed-form laws, indicating a degree of combinatorial rigidity unusual among Hofstadter-type recursions.

The first objective of this paper is to develop the combinatorial structure underlying these frequency laws. We prove that the frequencies are organized by a recursive dyadic lifting mechanism arising from plateau propagation in Cloître's binary step-word decomposition.

The second objective is to study deviations from exact dyadic scaling. For this purpose we introduce the renormalized defect $R(n) = Q(2n) - 2Q(n)$, which measures the failure of the relation $Q(2n) = 2Q(n)$.

Our main structural result is that the recursion admits an exact reformulation in terms of a single auxiliary clock process $t_1(n) = n - Q(n - 1)$, satisfying a closed delayed recursion. Using this reformulation, we prove the exact identity

$$R(n) = 2t_1(n + 1) - t_1(2n + 1) - 1,$$

which expresses the dyadic defects entirely through the clock dynamics.

The principal exact results obtained in this paper are:

- the dyadic value-frequency law:

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\};$$

- the dyadic plateau lifting mechanism for the excesses $F(s) - 3$;
- the dyadic hitting-time endpoint identity $T(2^k) = 2^{k+2} - k - 2$;
- reduction of the recursion to a single clock process and a closed delayed recursion for $t_1(n)$;
- the exact defect identity $R(n) = 2t_1(n+1) - t_1(2n+1) - 1$.

The numerical data further suggest coherent defect patterns under dyadic rescaling. In particular, normalized defect profiles plotted against $\log_2 n$ display possible logarithmic-scale organization. While these observations remain partly heuristic, they indicate that the recursion possesses a non-trivial recursive scaling structure beyond its coarse asymptotic growth behavior.

The present work should therefore be viewed as complementary to recent asymptotic investigations of the sequence. In particular, Cloître [2] proved that the recursion is globally well-defined and satisfies

$$Q(n) = \frac{n}{2} + O\left(\frac{n}{\sqrt{\log n}}\right).$$

Thus the coarse asymptotic growth problem is now resolved. The emphasis of the present paper is different: rather than studying only average growth, we investigate the finer dyadic fluctuation geometry generated by the recursion.

More broadly, the results suggest that certain nested recursions may exhibit persistent interactions between dyadic scaling, frequency structure, and logarithmic oscillations. The emergence of such phenomena from an elementary self-referential recursion appears to raise several new questions concerning the interaction between combinatorial recursion, asymptotic growth, and discrete scaling behavior.

The paper is organized as follows. Section 2 introduces basic notation and preliminary properties of the sequence. Section 3 proves the explicit dyadic frequency law. Section 4 develops the clock reduction and defect identity. Section 5 presents numerical evidence for logarithmic-scale organization of the defect profiles. Section 6 discusses scaling heuristics and asymptotic interpretation. Section 7 collects open problems and possible extensions. An appendix on alternative seeds illustrates that the canonical orbit is part of a broader seed-dependent dynamical landscape. Finally, we emphasize that the explicit frequency identities and the clock reduction formula are exact, whereas the scaling interpretations and log-periodic observations remain partly heuristic and numerical in nature.

2 Definitions and Preliminary Properties

2.1 Definition of the perturbed Hofstadter recursion

We study the integer sequence $Q(n)$ defined by $Q(1) = Q(2) = 1$ and

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n, \quad n \geq 3.$$

The sequence was introduced experimentally by the author as a perturbation of Hofstadter's classical Q -recursion [5]. The additional alternating term $(-1)^n$ significantly changes the global behavior of the recursion while preserving its nested self-referential structure.

The first values are

$$1, 1, 1, 3, 3, 3, 5, 5, 5, 7, 7, 9, 7, 9, 7, 11, 9, 11, 11, 11, \dots,$$

suggesting approximately linear growth with persistent local repetition patterns. Numerical computations further indicate regular binary-scaled patterns in both the value frequencies and the fluctuation profiles.

As in many Hofstadter-type systems, the recursive arguments $n - Q(n - 1)$ and $n - Q(n - 2)$ depend on the entire previous history of the sequence. Nevertheless, the present recursion seems to have more regularity than the classical Hofstadter sequence.

Since only odd values occur, it will be convenient to parameterize the values as $2m - 1$, $m \geq 1$. The corresponding value-frequency function will be introduced formally in Subsection 2.5. One of the main goals of the paper is to prove an explicit dyadic formula for these frequencies. In Section 3 we prove that, for every $k \geq 0$,

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

We also define the renormalized defect sequence $R(n) := Q(2n) - 2Q(n)$, which measures deviations from exact dyadic scaling. As will become apparent later, the quantities $R(n)$ encode much of the hidden fluctuation structure of the recursion.

2.2 Basic properties, parity, and first values

We begin with several elementary observations concerning the sequence.

Lemma 2.1 (Parity). *For all $n \geq 1$, the value $Q(n)$ is odd.*

Proof. The initial values $Q(1) = Q(2) = 1$ are odd. If all previous values are odd, then

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n$$

is the sum of two odd integers and either 1 or -1 . Hence $Q(n)$ is again odd. $\square$

The first values of the sequence are

$$1, 1, 1, 3, 3, 3, 5, 5, 5, 7, 7, 9, 7, 9, 7, 11, 9, 11, 11, 11, \dots$$

Thus only odd values occur. This motivates the parametrization $v = 2m - 1$, $m \geq 1$, which will be used throughout the paper. The corresponding value-frequency function is defined in Subsection 2.5.

The initial data already display several striking regularities. In particular, the sequence exhibits persistent local repetition structures together with highly nontrivial fluctuations around its dominant linear trend. Unlike the classical Hofstadter sequence, the present recursion appears numerically far more organized and stable.

Figure 1 shows the sequence up to $n = 4096$. It illustrates the fluctuation structure around the dominant linear growth law $Q(n) \sim n/2$, together with large-scale oscillatory deviations and repeated dyadic patterns.

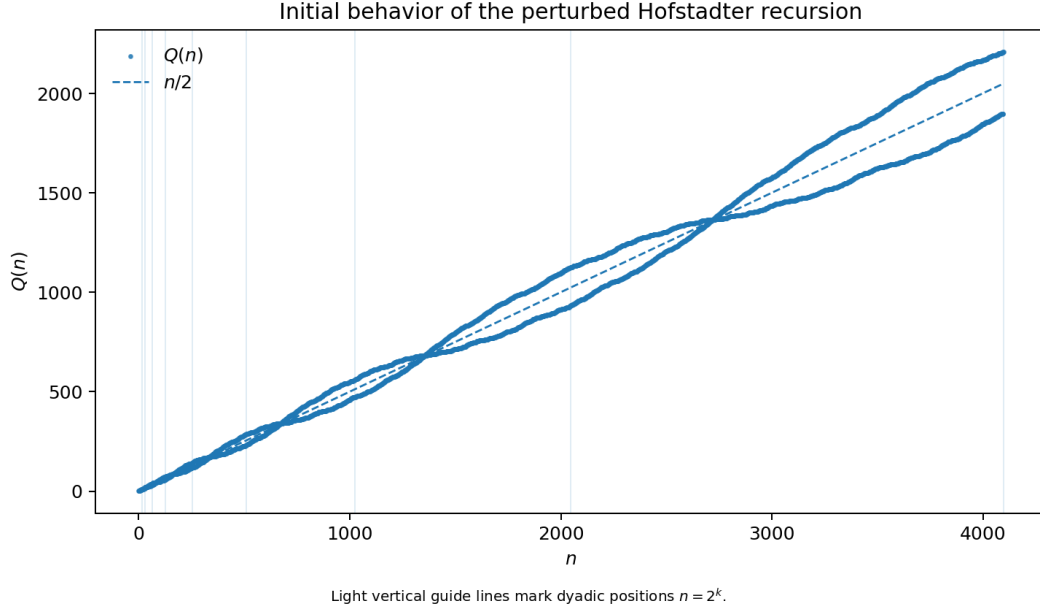


Figure 1: Initial values of the perturbed Hofstadter recursion up to $n = 4096$. The dashed line indicates the reference growth law $Q(n) = n/2$. Light vertical guide lines mark dyadic positions $n = 2^k$. The sequence exhibits approximately linear growth together with visible large-scale fluctuation patterns.

2.3 Safety margin

The recursive arguments $n - Q(n - 1)$ and $n - Q(n - 2)$ determine whether the recursion remains well-defined. We therefore define the safety margin

$$S(n) := n - \max\{Q(n - 1), Q(n - 2)\}.$$

Thus $S(n) > 0$ guarantees that both recursive evaluations remain valid. Empirically, the safety margin grows approximately linearly, $S(n) \approx n/2$. This indicates that the recursion remains well separated from the singular boundary and that the recursive evaluations typically occur near scale $n/2$.

Figure 2 shows the behavior of $S(n)$ up to $n = 4096$.

2.4 Monotonicity and slow growth

Many nested recursions exhibit slow-growth behavior, where consecutive values differ only by small amounts. Numerical experiments strongly suggest that the present sequence shares this property.

Conjecture 2.2. For all $n \geq 1$, $Q(n + 1) - Q(n) \in \{-2, 0, 2\}$.

Since all values of $Q(n)$ are odd by Lemma 2.1, only even increments can occur. Extensive computations support the conjecture up to at least $n = 10^7$. We write $D(n) := Q(n + 1) - Q(n)$ for the first-difference sequence; Figure 3 shows $D(n)$ up to $n = 4096$.

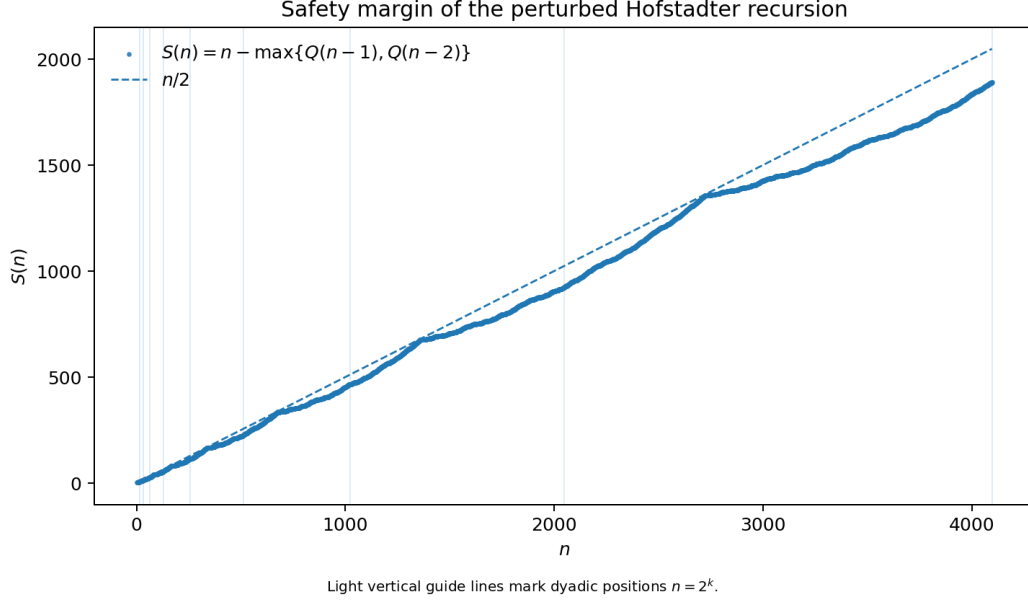


Figure 2: Safety margin $S(n) = n - \max\{Q(n-1), Q(n-2)\}$ for the perturbed Hofstadter recursion up to $n = 4096$. The dashed line indicates the reference growth law $S(n) = n/2$. Light vertical guide lines mark dyadic positions $n = 2^k$.

2.5 Auxiliary sequences and notation

We introduce several auxiliary quantities used throughout the paper. Since all values of $Q(n)$ are odd, we parameterize them as $2m - 1$, $m \geq 1$. The value-frequency function is

$$F(m) := \#\{n \geq 1 : Q(n) = 2m - 1\},$$

so that $F(m)$ counts occurrences of the odd value $2m - 1$ in the sequence. The corresponding cumulative counting function is $C(m) := \sum_{s \leq m} F(s)$, equivalently $C(m) = \#\{n \geq 1 : Q(n) \leq 2m - 1\}$.

A central role is played by the renormalized defect sequence $R(n) := Q(2n) - 2Q(n)$, which measures deviations from exact dyadic scaling. We also define the centered fluctuation function $E(n) := Q(n) - n/2$, which isolates deviations from the dominant linear growth term. Later sections introduce a self-referential clock dynamics $t_1(n)$ satisfying

$$R(n) = 2t_1(n+1) - t_1(2n+1) - 1.$$

Throughout the paper, logarithms are taken to base 2 unless otherwise specified.

2.6 Dyadic blocks and renormalized fluctuations

A recurring feature of the numerical data is the appearance of organized structures inside the dyadic intervals $B_k := [2^k, 2^{k+1})$, $k \geq 0$. We refer to these intervals as *dyadic blocks*.

Within each dyadic block B_k , the centered fluctuations exhibit reproducible large-scale profiles. After rescaling, successive blocks display visible geometric alignment, suggesting that the

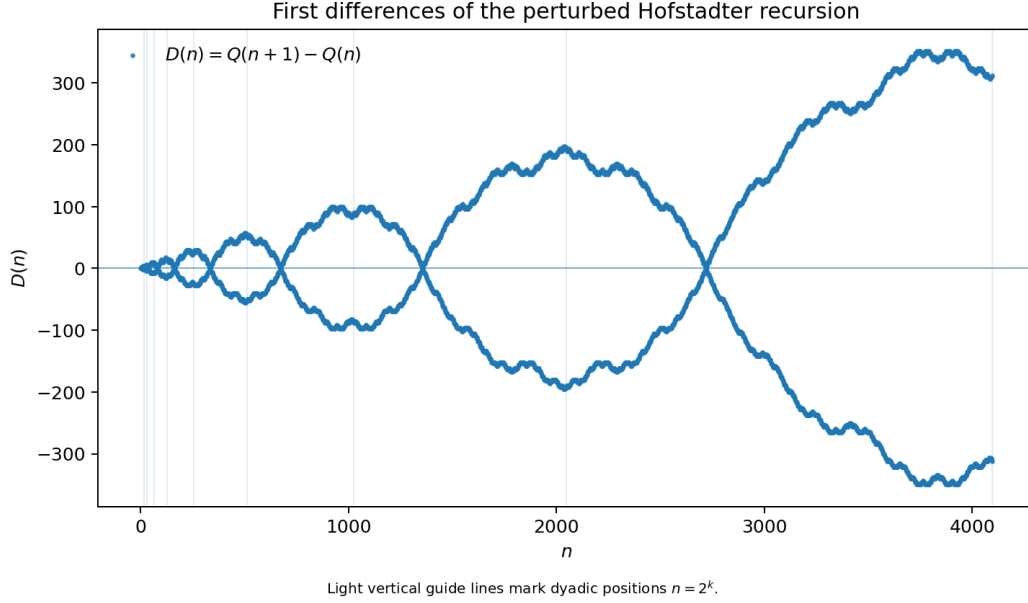


Figure 3: First differences $D(n) = Q(n+1) - Q(n)$ for the perturbed Hofstadter recursion up to $n = 4096$. The differences remain bounded and exhibit organized structures across dyadic scales. Light vertical guide lines mark dyadic positions $n = 2^k$.

fluctuation dynamics are not independent across scales. To study these effects, we use the centered fluctuation function $E(n) = Q(n) - n/2$ and, for each dyadic scale k , define

$$E_k(m) := Q(2^k m) - 2^{k-1} m.$$

The numerical behavior of these profiles motivates the renormalized defect analysis developed later in the paper. The detailed numerical collapse of the dyadic fluctuation profiles is presented in Section 5.

3 Explicit Frequency Law

3.1 Numerical motivation and statement of the law

One of the most remarkable features of the perturbed Hofstadter recursion is the emergence of highly regular value frequencies despite the nonlinear self-referential nature of the recurrence.

Recall from Subsection 2.5 that, since all values of $Q(n)$ are odd, we write them as $2m - 1$, $m \geq 1$, and define

$$F(m) := \#\{n \geq 1 : Q(n) = 2m - 1\}.$$

The first computed values are

$$(F(m))_{m \geq 1} = (3, 3, 4, 3, 3, 5, 4, 3, 3, 4, 6, 5, 3, 3, 4, 3, 3, \dots).$$

Thus, for instance, $F(1) = 3$, $F(2) = 3$, $F(3) = 4$, $F(4) = 3$, $F(5) = 3$, $F(6) = 5$, $F(7) = 4$, and $F(8) = 3$. Already these initial values reveal a nontrivial arithmetic pattern: the frequencies are not distributed randomly, but exhibit a rigid dyadic organization.

To visualize the distribution, Figure 4 shows the values of $F(m)$ and the running maximum $M(m) = \max_{1 \leq j \leq m} F(j)$ for $1 \leq m \leq 4000$.

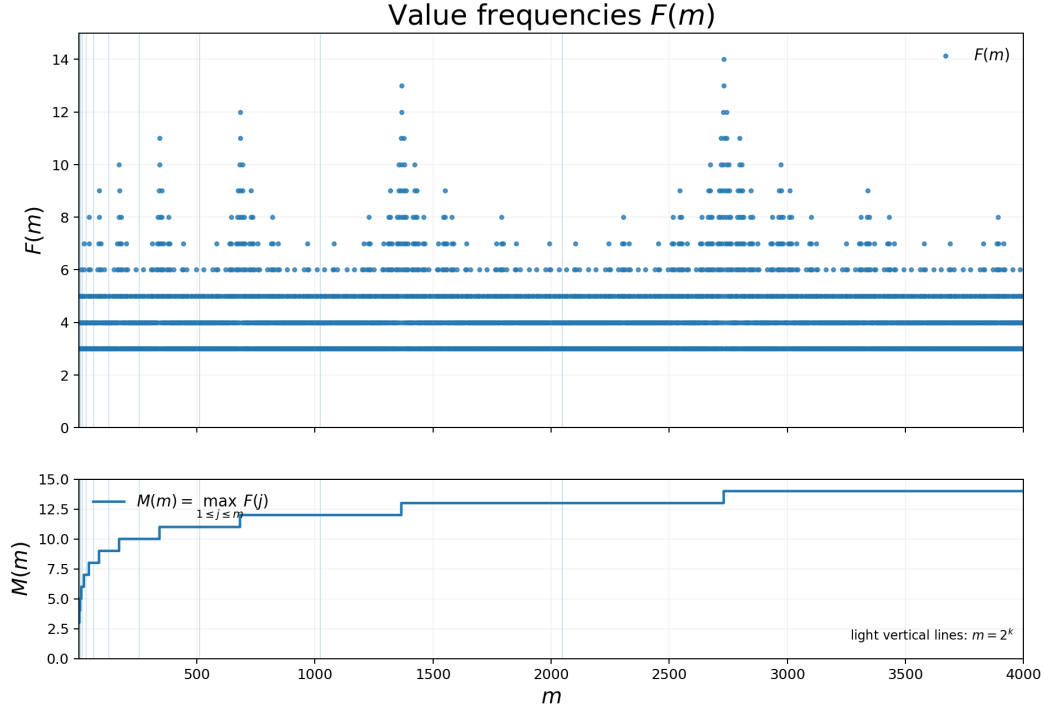


Figure 4: Frequency distribution $F(m) = |\{n \geq 1 : Q(n) = 2m - 1\}|$ for the perturbed Hofstadter recursion. The frequencies exhibit regular dyadic organization rather than random variation.

The horizontal band structure in Figure 4 suggests that only a small number of multiplicities occur at each scale, while the larger multiplicities appear at highly structured positions.

To make this structure explicit, we organize the frequencies according to dyadic value blocks $B_k := [2^k, 2^{k+1})$, $k \geq 0$. Inside each block B_k , the frequency profile displays repeated local structures that reappear at larger scales in modified form. The relevant arithmetic parameter is the dyadic valuation $\nu_2(j)$, the exponent of the highest power of 2 dividing j .

The main frequency theorem proved in this section is the following.

Theorem 3.1 (Dyadic frequency law). *For every $k \geq 0$,*

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\},$$

as multisets. Equivalently,

$$\#\{s \in B_k : F(s) = r\} = 2^{k-r+2}, \quad 3 \leq r \leq k+2,$$

and $\#\{s \in B_k : F(s) = k+3\} = 1$.

Thus the entire frequency distribution inside a dyadic block is governed by the classical ruler function $j \mapsto \nu_2(j)$, shifted upward by 3. In particular, the minimal frequency 3 occurs 2^{k-1} times

in B_k for $k \geq 1$, the next frequency 4 occurs 2^{k-2} times, and so on, with the largest value in the block occurring once.

For example, the theorem gives

$$\{F(s) : 1 \leq s < 2\} = \{3\}, \quad \{F(s) : 2 \leq s < 4\} = \{3, 4\}, \quad \{F(s) : 4 \leq s < 8\} = \{3, 3, 4, 5\},$$

and

$$\{F(s) : 8 \leq s < 16\} = \{3, 3, 3, 3, 4, 4, 5, 6\},$$

as multisets. These agree with the initial frequency data above.

To visualize the internal dyadic organization, Figure 5 overlays several normalized dyadic frequency blocks. For each block $B_k = [2^k, 2^{k+1})$, the horizontal coordinate is rescaled to unit length before plotting the corresponding frequencies.

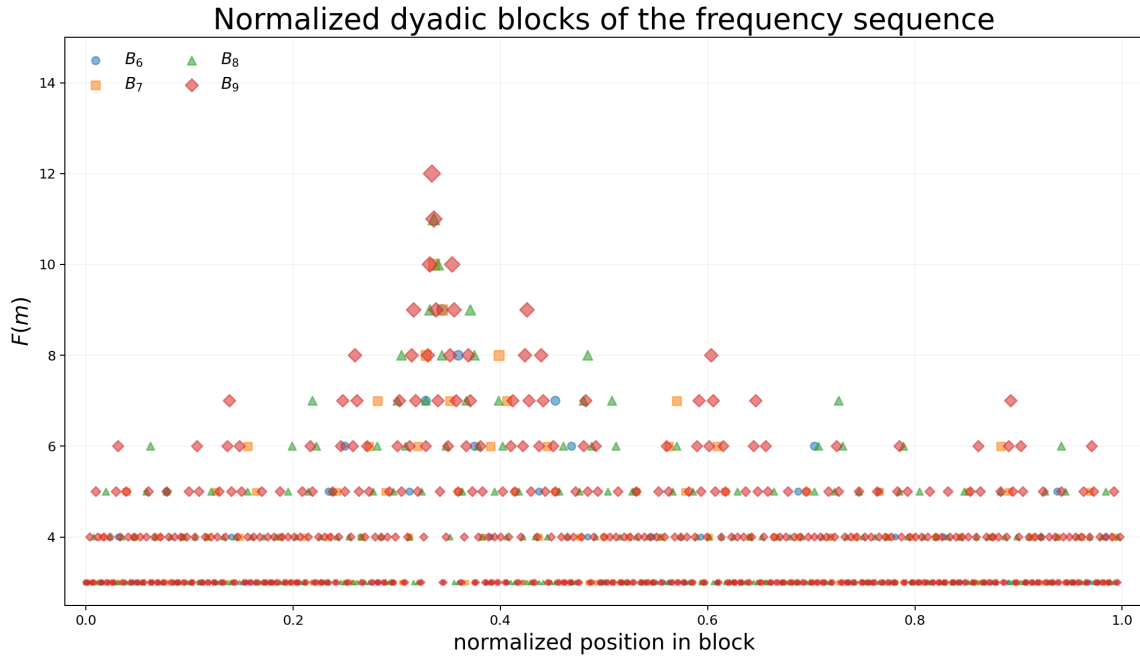


Figure 5: Normalized dyadic blocks of the frequency sequence $F(m)$. For each dyadic block $B_k = [2^k, 2^{k+1})$, the horizontal coordinate is normalized to unit length prior to overlaying the data. Large multiplicity peaks repeatedly occur near the same normalized positions across successive dyadic scales, indicating a persistent hierarchical organization of the frequency distribution.

Figure 5 illustrates the large-scale alignment of the frequency profiles under dyadic rescaling. The theorem explains this alignment: each block contains a shifted copy of the dyadic valuation pattern, and the high-frequency peaks occur precisely at positions corresponding to large powers of 2 in the local coordinate j .

Figure 6 gives a complementary numerical view of the same dyadic organization by tracking the normalized locations of dominant peaks across successive dyadic blocks.

The proof of Theorem 3.1 proceeds by translating the frequency problem into plateau lengths of Cloître’s monotone subsequences. The key steps are:

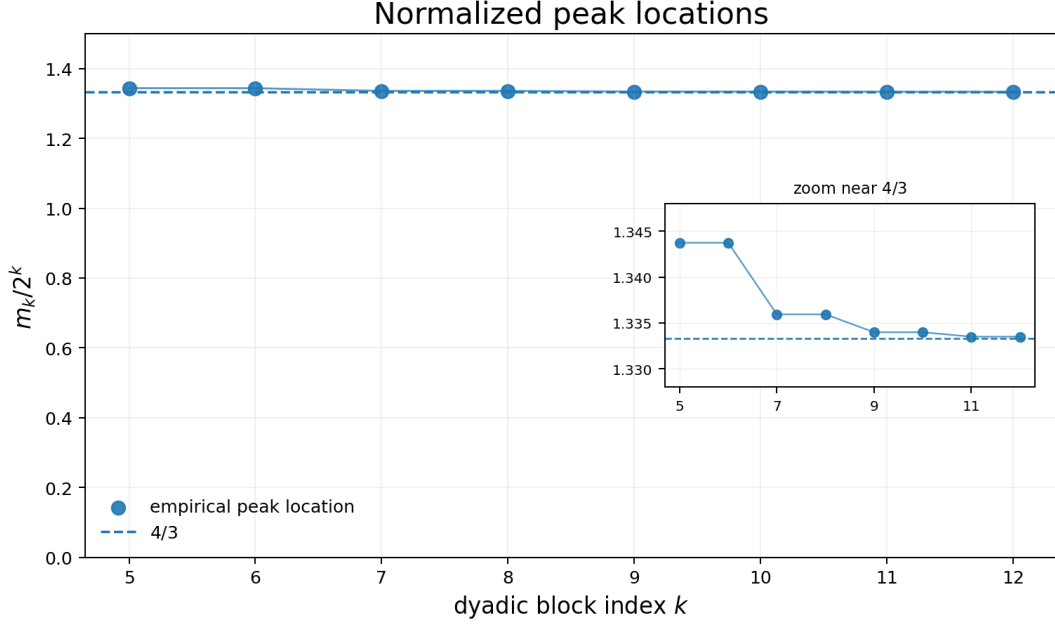


Figure 6: Normalized locations of dominant frequency peaks in successive dyadic blocks. The empirical ratios $m_k/2^k$ lie close to $4/3$ over the computed range, with the inset showing the residual deviations. This plot is included only as a numerical illustration of the dyadic organization.

- express $F(s)$ as the sum of the plateau lengths of the odd and even subsequences A and B ;
- identify these plateau lengths with zero-runs in binary step words;
- prove a dyadic endpoint identity for the corresponding hitting times;
- establish an induced rank-lifting mechanism for zero-runs across dyadic value blocks;
- use the resulting block-mass identity to force the exact dyadic excess lifting law.

The remaining subsections carry out these steps.

3.2 Cloître's monotone subsequences and plateau local times

We now pass from the original sequence $Q(n)$ to the parity subsequences introduced and analyzed by Cloître [2]. Since all values of $Q(n)$ are odd, define

$$A(m) := \frac{Q(2m-1)+1}{2}, \quad B(m) := \frac{Q(2m)+1}{2}.$$

Thus A records the odd-indexed values of Q , shifted and divided by 2, while B records the even-indexed values. Equivalently, $Q(2m-1) = 2A(m) - 1$ and $Q(2m) = 2B(m) - 1$.

Cloître proved that the increments

$$\Delta A(m) := A(m+1) - A(m), \quad \Delta B(m) := B(m+1) - B(m)$$

belong to $\{0, 1\}$. Thus A and B are nondecreasing integer-valued step functions with unit jumps.

For $s \geq 1$, define the plateau lengths

$$L_A(s) := \#\{m \geq 1 : A(m) = s\}, \quad L_B(s) := \#\{m \geq 1 : B(m) = s\}.$$

Since occurrences of the value $2s - 1$ in Q occur either at odd or even positions, the frequency decomposes as

$$F(s) = L_A(s) + L_B(s).$$

This gives the first bridge between the original recursion and the binary step-word structure: value frequencies of Q are exactly combined plateau local times of A and B .

We next express these plateau lengths in terms of zero-runs. For $X = A$ or $X = B$, let $\tau_X(s) := \min\{m : X(m) = s\}$, whenever s is attained. Since $\Delta X(m) \in \{0, 1\}$, the plateau at level s begins at $\tau_X(s)$ and ends just before $\tau_X(s + 1)$, hence

$$L_X(s) = \tau_X(s + 1) - \tau_X(s).$$

Equivalently, let $z_X(s)$ denote the length of the zero-run in the step word ΔX immediately following the first occurrence of the value s :

$$z_X(s) := \max\left\{\ell \geq 0 : \Delta X(\tau_X(s)) = \Delta X(\tau_X(s) + 1) = \cdots = \Delta X(\tau_X(s) + \ell - 1) = 0\right\}.$$

Then the plateau consists of the first occurrence of s , followed by exactly $z_X(s)$ additional repetitions, so $L_X(s) = 1 + z_X(s)$. Applying this to A and B , we obtain

$$F(s) = L_A(s) + L_B(s) = 2 + z_A(s) + z_B(s).$$

It will be useful to work with the excess frequency $H(s) := F(s) - 3$. In terms of zero-run lengths,

$$H(s) = z_A(s) + z_B(s) - 1.$$

Thus the minimal frequency $F(s) = 3$ corresponds to $z_A(s) + z_B(s) = 1$, while nonminimal frequencies correspond precisely to additional zero-run excess in the combined step words of A and B .

This reformulation reduces the dyadic frequency problem to a statement about zero-run propagation in Cloître's binary interleaving construction. In particular, to prove the dyadic frequency law it suffices to understand how the combined zero-run excess $z_A(s) + z_B(s) - 1$ is transported from one dyadic value block to the next.

3.3 Dyadic level-crossing words

The preceding subsection reduced the frequency problem to the study of zero-runs in the binary step words

$$\Delta A(m) = A(m + 1) - A(m), \quad \Delta B(m) = B(m + 1) - B(m).$$

We now restrict these step words to dyadic value intervals.

For $X = A$ or $X = B$, recall the hitting time $\tau_X(s) := \min\{m : X(m) = s\}$. Since X is nondecreasing and $\Delta X(m) \in \{0, 1\}$, the values of X are visited in consecutive order.

For each $k \geq 0$, define the dyadic level-crossing words

$$W_k^A := \Delta A[\tau_A(2^k), \tau_A(2^{k+1})], \quad W_k^B := \Delta B[\tau_B(2^k), \tau_B(2^{k+1})].$$

Thus W_k^A is the portion of the step word ΔA during which A passes from level 2^k to level 2^{k+1} , and similarly for B . Equivalently, W_k^A and W_k^B record the plateau structure of A and B over the dyadic value block $B_k = [2^k, 2^{k+1})$.

Lemma 3.2 (Zero-runs project to dyadic value blocks). *Let $X = A$ or $X = B$. Then the zero-runs of the step word*

$$\Delta X[\tau_X(2^k), \tau_X(2^{k+1})]$$

project exactly onto the plateau values $2^k \leq s < 2^{k+1}$. More precisely, for every $s \in [2^k, 2^{k+1})$, the plateau of X at level s is fully contained in the level-crossing interval

$$[\tau_X(2^k), \tau_X(2^{k+1})],$$

and its length is $L_X(s) = 1 + z_X(s)$.

Proof. Since X is nondecreasing and $\Delta X(m) \in \{0, 1\}$, the sequence X can only stay constant or increase by one. Hence, once X reaches 2^k , it must visit the values $2^k, 2^k + 1, \dots, 2^{k+1} - 1$ in consecutive order before reaching 2^{k+1} .

For a fixed $s \in [2^k, 2^{k+1})$, the plateau at level s begins at $\tau_X(s)$ and ends immediately before $\tau_X(s + 1)$. Thus the entire plateau interval is $[\tau_X(s), \tau_X(s + 1))$. Since $2^k \leq s < s + 1 \leq 2^{k+1}$, we have

$$[\tau_X(s), \tau_X(s + 1)) \subseteq [\tau_X(2^k), \tau_X(2^{k+1})].$$

During this plateau, the first occurrence of s is followed by exactly $z_X(s)$ zero increments in the step word ΔX . Hence $L_X(s) = 1 + z_X(s)$. $\square$

Consequently, the combined zero-run data of $W_k^A \cup W_k^B$ are exactly the frequency data inside the dyadic value block B_k . Indeed, for every $s \in B_k$,

$$F(s) = L_A(s) + L_B(s) = 2 + z_A(s) + z_B(s), \quad H(s) = F(s) - 3 = z_A(s) + z_B(s) - 1.$$

Thus the dyadic frequency law is equivalent to a statement about how the combined zero-run excess $z_A(s) + z_B(s) - 1$ is distributed among the zero-runs appearing in the two level-crossing words W_k^A and W_k^B .

The purpose of the next subsections is to show that Cloître's interleaving dynamics induce a rank-lifting mechanism between successive dyadic crossing words: zero-run excess in $W_k^A \cup W_k^B$ is lifted to zero-run excess in $W_{k+1}^A \cup W_{k+1}^B$ with one additional unit of length, while the remaining newly created zero-runs are minimal.

3.4 Dyadic endpoint theorem

We next record the dyadic hitting-time identity needed for the frequency proof. This is the point at which Cloître's interleaving structure enters in an essential way.

Recall that, for $X = A$ or $X = B$, $\tau_X(s) := \min\{m : X(m) = s\}$, and define the combined hitting-time function $T(s) := \tau_A(s) + \tau_B(s)$. Since

$$L_A(s) = \tau_A(s + 1) - \tau_A(s), \quad L_B(s) = \tau_B(s + 1) - \tau_B(s),$$

we have $F(s) = L_A(s) + L_B(s) = T(s+1) - T(s)$. Thus information about the values of $T(s)$ at dyadic points gives exact information about the total frequency mass inside dyadic value blocks.

Proposition 3.3 (Dyadic endpoint theorem). *For every $k \geq 0$,*

$$T(2^k) = 2^{k+2} - k - 2.$$

Proof. The proof is given in Appendix A; we briefly indicate the structure. Cloître's construction decomposes the step words of A and B into positive and negative arch words P_r and N_r , related by

$$N_r = \text{Interleave}(P_r[2:], P_r[: -1], 1), \quad P_{r+1} = \text{Interleave}([0, 0] \circ N_r \circ [1], [0] \circ N_r, 0).$$

For the dyadic endpoints 2^k , the relevant hitting times reduce to quarter-crossing times inside these binary words. For even dyadic exponents, $2^k = 4^{r+1}$, the hitting-time identity is equivalent to a first-passage relation in the positive word P_r . For odd dyadic exponents, $2^k = 2 \cdot 4^{r+1}$, it is equivalent to the corresponding first-passage relation in the negative word N_r . These two first-passage relations are proved in Appendix A by using the height function $h_W(t) = Z_W(t) - O_W(t)$, height additivity of the interleaving operation, and the quarter-head alignment identities for the two tapes. The resulting quarter-crossing identities imply $T(2^k) = 2^{k+2} - k - 2$ for all $k \geq 0$. $\square$

Corollary 3.4 (Dyadic block mass). *For every $k \geq 0$,*

$$\sum_{s=2^k}^{2^{k+1}-1} F(s) = 4 \cdot 2^k - 1.$$

Consequently, since $H(s) = F(s) - 3$,

$$\sum_{s=2^k}^{2^{k+1}-1} H(s) = 2^k - 1.$$

Proof. Using $F(s) = T(s+1) - T(s)$, we obtain

$$\sum_{s=2^k}^{2^{k+1}-1} F(s) = T(2^{k+1}) - T(2^k).$$

By Proposition 3.3,

$$T(2^{k+1}) = 2^{k+3} - (k+1) - 2, \quad T(2^k) = 2^{k+2} - k - 2.$$

Therefore

$$T(2^{k+1}) - T(2^k) = (2^{k+3} - k - 3) - (2^{k+2} - k - 2) = 2^{k+2} - 1 = 4 \cdot 2^k - 1.$$

This proves the first claim. Since $B_k = [2^k, 2^{k+1})$ has cardinality 2^k , we have

$$\sum_{s \in B_k} H(s) = \sum_{s \in B_k} (F(s) - 3) = (4 \cdot 2^k - 1) - 3 \cdot 2^k = 2^k - 1.$$

$\square$

The corollary gives the exact total amount of excess frequency available in each dyadic block. In the following subsection, this mass identity will be combined with an induced zero-run lifting mechanism. The key point is that once an injective lift of excess from B_k into B_{k+1} is established, the block mass identity forces the lift to be exact and leaves no room for additional nonminimal excess outside the lifted image.

3.5 Induced rank-lift and mass closure

We now prove the dyadic excess lifting law. The argument has two parts: first, Cloître's interleaving mechanism induces an injective lift of zero-run excess from one dyadic value block to the next; second, the dyadic block mass identity from Corollary 3.4 forces this lift to be exact.

Recall that $B_k = [2^k, 2^{k+1})$, $H(s) := F(s) - 3$, and, by the plateau local-time formula,

$$H(s) = z_A(s) + z_B(s) - 1.$$

We prove the induced rank-lift and the excess lifting theorem simultaneously by induction on k . For $k \geq 1$, the induction hypothesis implies that every nonminimal excess in B_k has a unique excess-carrying source from B_{k-1} . The case $k = 0$ is treated separately as the base case. This uniqueness is supplied by the global run-accounting property of the interleaving machine.

Definition 3.5 (Combined zero-run package). For $t \in B_k$, the combined zero-run package at level t is

$$\mathcal{Z}_k(t) := (z_A(t), z_B(t)).$$

Its excess is $e(\mathcal{Z}_k(t)) := z_A(t) + z_B(t) - 1 = H(t)$. A package is called nonminimal if $H(t) > 0$.

Lemma 3.6 (Local return rule). *Let $Z = \text{Interleave}(X, Y, b)$, with $b \in \{0, 1\}$. Suppose the machine reads a zero from the Y -tape and thereby returns to state 0. If the X -head is at the beginning of a zero-run 0^ℓ in X , then the output contains a zero-run of length $\ell + 1$. If the X -head is not at the beginning of a zero-run, the zero read from Y contributes only an isolated zero. Moreover, since both tape heads move monotonically, each zero-run of X can be lifted in this way at most once.*

The following proposition is the global form of the local return rule restricted to the dyadic crossing windows. It is the only point in the frequency proof where the local interleaving dynamics are used to produce a map between consecutive dyadic value blocks.

Lemma 3.7 (Run accounting for the interleaving machine). *Let*

$$Z = \text{Interleave}(X, Y, b), \quad b \in \{0, 1\},$$

and let i_t and j_t denote the numbers of symbols consumed from X and Y , respectively, after t output symbols have been produced. Assume that, in the range under consideration, every nonisolated zero-run considered below is produced before any forced read caused by exhaustion of one tape.

Let $J = [p, q)$ be a maximal zero-run of Z with length $q - p \geq 2$. Then J has a unique source zero-run in the X -tape in the following sense.

If the first zero of J is read from the X -tape, then the source is the unique maximal zero-run of X beginning at position i_p . In this case J is exactly the image of this source run.

If the first zero of J is read from the Y -tape, then this first zero returns the machine to state 0, and the remaining zeros of J are read from the X -tape. In this case the source is the unique

maximal zero-run of X beginning at position i_{p+1} , and the length of J is one larger than the length of this source run.

Consequently:

1. every nonisolated zero-run of Z has a unique source zero-run in the X -tape;
2. two distinct source zero-runs of X cannot produce the same nonisolated zero-run of Z ;
3. the correspondence between source zero-runs and their nonisolated target zero-runs is order-preserving.

Moreover, let $I_X = [a, b)$ be an interval of the X -tape and let $I_Z = [u, v)$ be an output interval such that the X -head enters I_X at time u , leaves I_X at time v , and every zero-run of X intersecting I_X is either fully contained in I_X or disjoint from it. Then no nonisolated zero-run generated by a source zero-run in I_X is lost when the interleaving is restricted to I_Z . Conversely, every nonisolated zero-run of Z contained in I_Z has its source zero-run in I_X .

Proof. During the interleaving process the current state is always the last emitted bit. Hence, after a zero has been emitted, the machine is in state 0 and the next prescribed read is from the X -tape.

Let $J = [p, q)$ be a maximal zero-run of Z with $q - p \geq 2$. If the symbol $Z(p)$ is read from the X -tape, then the machine remains in state 0 after emitting $Z(p) = 0$. Therefore all subsequent symbols of the same zero-run are also read from the X -tape, until the first symbol 1 is encountered in X . Thus J is precisely the image of the maximal zero-run of X beginning at i_p .

If $Z(p)$ is read from the Y -tape, then immediately before time p the machine is in state 1. Since $Z(p) = 0$, this read returns the machine to state 0. Because J has length at least 2, the next symbol $Z(p + 1)$ exists and is again zero. It is therefore read from the X -tape. From that point onward, as long as zeros continue to be emitted, the state remains 0, so the machine continues reading consecutively from the X -tape. Hence the remaining part $Z[p + 1 : q)$ is exactly the image of the maximal zero-run of X beginning at i_{p+1} . In this case the initial zero from the Y -tape contributes one additional zero, so the target run has length one more than its source run.

This proves existence of a source zero-run for every nonisolated zero-run of Z . The source is unique because the X -head moves monotonically and a maximal zero-run of X is separated from the next one by a symbol 1. Once this separating 1 is read, the output zero-run terminates, so two different zero-runs of X cannot merge into a single zero-run of Z .

Similarly, two distinct source zero-runs of X cannot produce the same target run: the X -head visits them in their natural order, and between two distinct source zero-runs it must read a 1, which separates the corresponding target runs. Hence the correspondence is injective.

The same monotonicity also proves order preservation. If one source zero-run of X lies strictly before another, then the X -head encounters the first before the second. Since output time is increasing along the run of the machine, the target zero-run generated by the first source occurs before the target zero-run generated by the second source.

Finally, suppose $I_X = [a, b)$ and $I_Z = [u, v)$ satisfy the stated window compatibility assumptions. Because the X -head enters I_X at time u and leaves it at time v , every source zero-run contained in I_X is encountered completely during the output interval I_Z . Since source zero-runs are not cut by the boundary of I_X , the preceding argument shows that their target zero-runs are also contained in I_Z . Conversely, if a nonisolated zero-run of Z is contained in I_Z , then its source is the X -zero-run being traversed by the X -head during that output interval, hence lies in I_X . This proves that the restriction to the window loses no relevant nonisolated zero-runs. $\square$

Lemma 3.8 (Compatibility of crossing windows). *For $X = A$ or $X = B$, the crossing window*

$$\Delta X[\tau_X(2^k), \tau_X(2^{k+1}))$$

does not cut any plateau zero-run associated with a value $s \in B_k = [2^k, 2^{k+1})$. Hence every zero-run contributing to $z_X(s)$, $s \in B_k$, is fully contained in the corresponding crossing window.

Proof. By the zero-run projection lemma, the plateau of X at level $s \in B_k$ is exactly

$$[\tau_X(s), \tau_X(s+1)).$$

Since $2^k \leq s < s+1 \leq 2^{k+1}$, this interval is contained in

$$[\tau_X(2^k), \tau_X(2^{k+1})).$$

Thus no zero-run associated with a plateau value in B_k is cut by the boundary of the crossing window. $\square$

Proposition 3.9 (Induced crossing lift). *For every $k \geq 0$, the interleaving laws of Cloître induce an order-preserving raw lift*

$$\tilde{\Lambda}_k : B_k \longrightarrow \mathbb{N}$$

from combined zero-run packages in $W_k^A \cup W_k^B$ to zero-run packages in the ambient target crossing structure. Each lifted package contains one zero-run whose length is larger by one than the corresponding source zero-run.

Moreover, every nonisolated new zero-run in the target crossing structure has a unique source zero-run in the preceding crossing structure, and the order of source packages is preserved. In particular, whenever $\tilde{\Lambda}_k(t) \in B_{k+1}$, one has

$$H(\tilde{\Lambda}_k(t)) \geq H(t) + 1.$$

Proof. Apply Lemma 3.7 to the two interleaving laws

$$N_r = \text{Interleave}(P_r[2:], P_r[: -1], 1), \quad P_{r+1} = \text{Interleave}([0, 0] \circ N_r \circ [1], [0] \circ N_r, 0).$$

Inside each dyadic crossing window, the tape heads move monotonically and no relevant source zero-run is cut by the boundary of the crossing interval. Thus the window compatibility condition in Lemma 3.7 applies.

For a combined package $Z_k(t) = (z_A(t), z_B(t))$, the local return rule lifts one of its zero-run components by one unit whenever the machine returns from the Y -tape to state 0 with the X -head at the beginning of that zero-run. Lemma 3.7 gives uniqueness of the source, injectivity, and preservation of order. This defines the raw lift $\tilde{\Lambda}_k$. The image is not yet asserted to lie in B_{k+1} ; this is proved in Lemmas 3.11 and 3.12. $\square$

For each $t \in B_k$, Proposition 3.9 assigns to the combined zero-run package $Z_k(t)$ a unique raw lifted package in the ambient target structure. We denote by $\tilde{\Lambda}_k(t)$ the value supporting this package.

Lemma 3.10 (Unique source for nonminimal excess). *Let $k \geq 1$. Assume that the dyadic excess lifting identity is known up to rank $k-1$. Then every nonminimal package in B_k lies in the image of the preceding lift $\Lambda_{k-1} : B_{k-1} \rightarrow B_k$ and has a unique preimage under this lift.*

Proof. By the dyadic excess lifting identity at rank $k - 1$,

$$\{H(s) : s \in B_k\} = \{0^{(2^{k-1})}\} \sqcup \{H(t) + 1 : t \in B_{k-1}\}.$$

Moreover, the construction of the lift at rank $k - 1$ gives an injective map $\Lambda_{k-1} : B_{k-1} \rightarrow B_k$ such that $H(\Lambda_{k-1}(t)) = H(t) + 1$. Therefore every value $s \in B_k$ with $H(s) > 0$ must lie in the image of Λ_{k-1} . Since Λ_{k-1} is injective, this preimage is unique. $\square$

Lemma 3.11 (No left leakage). *Let $k \geq 1$, and assume that the dyadic excess lifting identity is known up to rank $k - 1$. Then, for every $t \in B_k$, the raw lifted value $\tilde{\Lambda}_k(t)$ cannot lie in B_k . Equivalently, $\tilde{\Lambda}_k(t) \geq 2^{k+1}$.*

Proof. Suppose, to the contrary, that for some $t \in B_k$,

$$\tilde{\Lambda}_k(t) < 2^{k+1}.$$

Since the raw lift moves forward through the dyadic crossing structure, the lifted value would have to lie in B_k .

By construction of the raw lift, the target package contains a lifted nonisolated zero-run and therefore has positive excess. Hence $\tilde{\Lambda}_k(t)$ would be a nonminimal package in B_k .

For $k \geq 1$, Lemma 3.10 implies that every nonminimal package in B_k already has a unique excess-carrying source from B_{k-1} . By Lemma 3.7, every nonisolated lifted zero-run has a unique source. Hence the same nonminimal zero-run cannot also have $t \in B_k$ as a second source. This contradiction proves the claim. $\square$

Lemma 3.12 (No right leakage). *The raw lifted image cannot leave B_{k+1} to the right. Equivalently, for every $t \in B_k$, $\tilde{\Lambda}_k(t) < 2^{k+2}$.*

Proof. Suppose that right leakage occurs. Since the raw lift is order-preserving, there is a first value $t_0 \in B_k$ such that

$$\tilde{\Lambda}_k(t_0) \geq 2^{k+2}.$$

Then only the values $t < t_0$ can contribute lifted excess inside B_{k+1} . Hence the total excess mass contributed inside B_{k+1} by lifted packages is strictly smaller than

$$\sum_{t \in B_k} (H(t) + 1).$$

By Corollary 3.4,

$$\sum_{t \in B_k} H(t) = 2^k - 1,$$

and therefore

$$\sum_{t \in B_k} (H(t) + 1) = (2^k - 1) + 2^k = 2^{k+1} - 1.$$

On the other hand, the total available excess mass in B_{k+1} is also

$$\sum_{s \in B_{k+1}} H(s) = 2^{k+1} - 1.$$

The newly created zero-runs not arising from lifted old runs are isolated and contribute no excess. Thus there is no additional source that could compensate for the lost lifted mass. This contradiction excludes right leakage. $\square$

Lemma 3.13 (Induced rank-lift at rank k). *Assume the dyadic excess lifting identity holds up to rank $k - 1$ if $k \geq 1$, with the case $k = 0$ handled directly. Then there exists an injective map*

$$\Lambda_k : B_k \longrightarrow B_{k+1}$$

such that

$$H(\Lambda_k(t)) \geq H(t) + 1$$

for every $t \in B_k$.

Proof. Consider the dyadic level-crossing words

$$W_k^A = \Delta A[\tau_A(2^k), \tau_A(2^{k+1})], \quad W_k^B = \Delta B[\tau_B(2^k), \tau_B(2^{k+1})].$$

By the zero-run projection lemma, the combined zero-run data of $W_k^A \cup W_k^B$ are exactly the plateau data for the values $t \in B_k$.

By Lemma 3.6 and Proposition 3.9, the local zero-run propagation inside the crossing window produces the raw lift

$$t \mapsto \tilde{\Lambda}_k(t).$$

The interleaving heads move monotonically through the input tapes. Hence the order of the lifted packages is the same as the order of the original levels:

$$t_1 < t_2 \implies \tilde{\Lambda}_k(t_1) < \tilde{\Lambda}_k(t_2).$$

Thus the raw lift $t \mapsto \tilde{\Lambda}_k(t)$ is injective.

By Lemma 3.11 and Lemma 3.12, no raw lifted value can lie outside B_{k+1} . Therefore

$$\tilde{\Lambda}_k(B_k) \subseteq B_{k+1}.$$

We define

$$\Lambda_k := \tilde{\Lambda}_k.$$

Finally, by construction at least one zero-run in the package associated with t is lifted with its length increased by one. Hence

$$z_A(\Lambda_k(t)) + z_B(\Lambda_k(t)) \geq z_A(t) + z_B(t) + 1.$$

Using

$$H(s) = z_A(s) + z_B(s) - 1,$$

we obtain

$$H(\Lambda_k(t)) \geq H(t) + 1.$$

This proves the lemma. □

Theorem 3.14 (Dyadic excess lifting). *For every $k \geq 0$,*

$$\{H(s) : s \in B_{k+1}\} = \{0^{(2^k)}\} \sqcup \{H(t) + 1 : t \in B_k\},$$

as multisets.

Proof. We argue by induction on k . The initial case $k = 0$ follows directly from $B_0 = \{1\}$, $H(1) = 0$, $B_1 = \{2, 3\}$, $F(2) = 3$, and $F(3) = 4$. Thus $H(2) = 0$, $H(3) = 1$, and

$$\{H(s) : s \in B_1\} = \{0, 1\} = \{0^{(1)}\} \sqcup \{H(1) + 1\}.$$

Now let $k \geq 1$, and assume that the excess lifting identity is known up to rank $k - 1$. By Lemma 3.13, there is an injective map $\Lambda_k : B_k \rightarrow B_{k+1}$ such that $H(\Lambda_k(t)) \geq H(t) + 1$ for every $t \in B_k$.

Therefore

$$\sum_{t \in B_k} H(\Lambda_k(t)) \geq \sum_{t \in B_k} (H(t) + 1).$$

Using Corollary 3.4, we get

$$\sum_{t \in B_k} (H(t) + 1) = (2^k - 1) + 2^k = 2^{k+1} - 1.$$

But the total excess mass in the whole block B_{k+1} is also

$$\sum_{s \in B_{k+1}} H(s) = 2^{k+1} - 1.$$

Since all excesses are nonnegative, equality must hold throughout. Hence

$$H(\Lambda_k(t)) = H(t) + 1$$

for every $t \in B_k$, and every value

$$s \in B_{k+1} \setminus \Lambda_k(B_k)$$

has

$$H(s) = 0.$$

Since $|B_k| = 2^k$, the image $\Lambda_k(B_k)$ has 2^k elements, and the complement in B_{k+1} also has 2^k elements. Therefore, as multisets,

$$\{H(s) : s \in B_{k+1}\} = \{0^{(2^k)}\} \sqcup \{H(t) + 1 : t \in B_k\}.$$

□

This theorem gives the exact dyadic plateau-lifting mechanism underlying the frequency law. It says that, from one dyadic value block to the next, half of the values are newly created with minimal frequency, while the other half inherit the previous excess frequency shifted upward by one.

3.6 Proof of the dyadic frequency law

We now complete the proof of Theorem 3.1. Recall that $H(s) = F(s) - 3$ and $B_k = [2^k, 2^{k+1})$. By Theorem 3.14, for every $k \geq 0$,

$$\{H(s) : s \in B_{k+1}\} = \{0^{(2^k)}\} \sqcup \{H(t) + 1 : t \in B_k\}.$$

We first prove the corresponding statement for the excesses.

Lemma 3.15 (Ruler-function law for the excesses). *For every $k \geq 0$,*

$$\{H(s) : s \in B_k\} = \{\nu_2(j) : 1 \leq j \leq 2^k\},$$

as multisets.

Proof. We argue by induction on k . For $k = 0$, $B_0 = [1, 2) = \{1\}$. Since the value $Q(n) = 1$ occurs three times, $F(1) = 3$, hence $H(1) = 0$. Also $\nu_2(1) = 0$. Thus

$$\{H(s) : s \in B_0\} = \{0\} = \{\nu_2(j) : 1 \leq j \leq 1\}.$$

Assume now that

$$\{H(s) : s \in B_k\} = \{\nu_2(j) : 1 \leq j \leq 2^k\}.$$

By Theorem 3.14 and the induction hypothesis,

$$\{H(s) : s \in B_{k+1}\} = \{0^{(2^k)}\} \sqcup \{1 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

On the other hand, the integers $1, \dots, 2^{k+1}$ split into 2^k odd integers, each with dyadic valuation 0, and the even integers $2j$, $1 \leq j \leq 2^k$, for which $\nu_2(2j) = 1 + \nu_2(j)$. Hence

$$\{\nu_2(j) : 1 \leq j \leq 2^{k+1}\} = \{0^{(2^k)}\} \sqcup \{1 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

Therefore

$$\{H(s) : s \in B_{k+1}\} = \{\nu_2(j) : 1 \leq j \leq 2^{k+1}\}.$$

This completes the induction. □

Since $F(s) = H(s) + 3$, the preceding lemma immediately gives

$$\{F(s) : s \in B_k\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

Equivalently, for every $k \geq 0$,

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\}.$$

It remains only to record the equivalent multiplicity form. For $0 \leq a < k$,

$$\#\{1 \leq j \leq 2^k : \nu_2(j) = a\} = 2^{k-a-1},$$

while $\nu_2(2^k) = k$. Since $F = 3 + \nu_2$, this means that, for $3 \leq r \leq k + 2$,

$$\#\{s \in B_k : F(s) = r\} = 2^{k-r+2},$$

except at the top value, where $\#\{s \in B_k : F(s) = k + 3\} = 1$. Thus B_k contains 2^{k-1} values of frequency 3, 2^{k-2} values of frequency 4, and so on, ending with one value of frequency $k + 2$ and one value of frequency $k + 3$. This is precisely the stated dyadic block frequency law.

3.7 Counting consequences

Let $C(m) := \sum_{s \leq m} F(s)$. By the dyadic frequency law, if $s = 2^k + j - 1$ with $1 \leq j \leq 2^k$, then $F(s) = 3 + \nu_2(j)$. Hence

$$\sum_{s=2^k}^{2^{k+1}-1} F(s) = \sum_{j=1}^{2^k} (3 + \nu_2(j)) = 3 \cdot 2^k + \sum_{j=1}^{2^k} \nu_2(j).$$

Since $\sum_{j=1}^{2^k} \nu_2(j) = 2^k - 1$, we obtain

$$\sum_{s=2^k}^{2^{k+1}-1} F(s) = 4 \cdot 2^k - 1.$$

Consequently,

$$C(2^k - 1) = \sum_{\ell=0}^{k-1} (4 \cdot 2^\ell - 1) = 4(2^k - 1) - k.$$

More generally, write $m = 2^k + r$, $0 \leq r < 2^k$, where $k = \lfloor \log_2 m \rfloor$. Then

$$C(m) = 4m - k - s_2(r + 1).$$

In particular, $C(m) = 4m + O(\log m)$. Thus the cumulative counting function has linear main term $4m$, with logarithmic dyadic fluctuations determined explicitly by the binary digit sum.

4 Dyadic Scaling Behavior and Clock Dynamics

4.1 The renormalized defect sequence

The numerical experiments presented in the previous sections suggest that the dominant linear growth $Q(n) \sim n/2$ does not fully capture the structure of the perturbed Hofstadter recursion. Although the coarse asymptotic behavior is comparatively simple, the residual fluctuations remain highly structured after dyadic rescaling.

To isolate these fluctuations, we introduce the renormalized defect sequence

$$R(n) := Q(2n) - 2Q(n).$$

The quantity $R(n)$ measures the deviation from exact dyadic scaling: if one had $Q(2n) = 2Q(n)$, then $R(n)$ would vanish identically. Instead, the numerical data indicate that $R(n)$ exhibits highly nontrivial oscillatory structure.

Equivalently, if $E(n) := Q(n) - n/2$ denotes the centered fluctuation function, then

$$R(n) = E(2n) - 2E(n).$$

Thus $R(n)$ measures the failure of the fluctuations themselves to obey exact dyadic self-similarity.

The first values are

$$(R(n))_{n \geq 1} = (-1, -1, -1, -1, -5, -1, -5, -1, -9, -1, -5, -1, \dots),$$

already indicating substantial arithmetic organization rather than random variation.

Figure 7 shows the renormalized defects up to $n = 4096$. The defect sequence isolates the nontrivial fluctuation dynamics hidden beneath the dominant linear growth term.

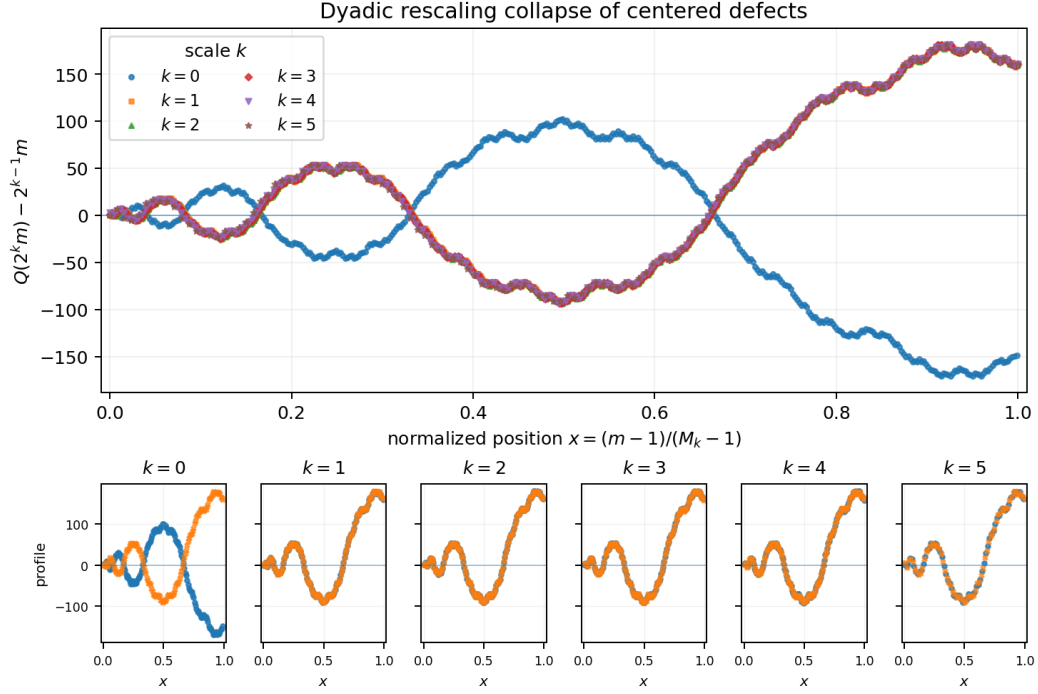


Figure 7: Renormalized defect sequence $R(n) = Q(2n) - 2Q(n)$ for the perturbed Hofstadter recursion up to $n = 4096$. The defects exhibit structured oscillatory behavior across dyadic scales. Light vertical guide lines mark dyadic positions $n = 2^k$.

4.2 Empirical properties of the defect sequence

The renormalized defects $R(n) = Q(2n) - 2Q(n)$ exhibit substantially more visible structure than the original sequence itself. After removal of the dominant linear growth term, the remaining fluctuations reveal persistent dyadic organization across multiple scales.

Several empirical properties emerge consistently throughout the numerical computations.

1. *Sublinear fluctuation growth.* The defects remain comparatively small relative to the scale n , consistent with Cloître's estimate $Q(n) = n/2 + O(n/\sqrt{\log n})$, which implies $R(n) = O(n/\sqrt{\log n})$.
2. *Persistent dyadic organization.* Inside successive dyadic windows $[2^k, 2^{k+1}]$, the defect profiles display similar large-scale shapes under dyadic rescaling. The fluctuations therefore appear highly structured rather than random.
3. *Band formation and binary structure.* The defects organize into visible bands correlated with parity and binary structure. Similar separation patterns already appeared earlier in the rescaled fluctuation profiles.
4. *Recursive propagation across scales.* The values $R(2n)$, $R(4n)$, and $R(8n)$ often exhibit closely related local behavior, suggesting that the fluctuation dynamics propagate recursively across dyadic scales.

These observations motivate the clock reduction developed in the next subsection.

4.3 Reduction to a single clock process

To describe the recursive index dynamics, define the auxiliary clock process

$$t_1(n) := n - Q(n - 1).$$

The recursive dynamics are governed by the shifted indices $n - Q(n - 1)$ and $n - Q(n - 2)$. At first sight these appear to be two independent recursive index processes; however, the second is completely determined by the first.

Proposition 4.1 (Reduction to a single clock process). *For every $n \geq 3$,*

$$n - Q(n - 2) = t_1(n - 1) + 1.$$

Proof. By definition, $t_1(n - 1) = (n - 1) - Q(n - 2)$. Rearranging immediately yields $n - Q(n - 2) = t_1(n - 1) + 1$. $\square$

Hence both recursive arguments may be expressed through the single process $t_1(n)$, and therefore

$$Q(n) = Q(t_1(n)) + Q(t_1(n - 1) + 1) + (-1)^n.$$

4.4 Reconstruction of the sequence from the clock

The reduction obtained in the previous subsection shows that the recursive index dynamics are governed by the single clock process $t_1(n) = n - Q(n - 1)$. We now show that the original sequence itself may be recovered directly from the clock dynamics. Thus the clock process does not merely encode auxiliary timing information: it completely determines the recursion.

Proposition 4.2 (Reconstruction from the clock process). *For every $m \geq 1$,*

$$Q(m) = m + 1 - t_1(m + 1).$$

Proof. By definition, $t_1(m + 1) = m + 1 - Q(m)$. Rearranging yields the claim. $\square$

Thus $Q(n)$ may be reconstructed directly from $t_1(n)$. Equivalently,

$$E(n) = 1 + \frac{n}{2} - t_1(n + 1).$$

4.5 Closed recursion for the clock process

The previous subsections showed that both the recursive arguments and the sequence values may be expressed through the clock process $t_1(n) = n - Q(n - 1)$. We now derive a closed delayed recursion satisfied by $t_1(n)$ itself.

Theorem 4.3 (Closed clock recursion). *For every $n \geq 3$,*

$$\begin{aligned} t_1(n + 1) &= n - 2 - t_1(n) - t_1(n - 1) \\ &\quad + t_1(t_1(n) + 1) + t_1(t_1(n - 1) + 2) - (-1)^n. \end{aligned}$$

Proof. Starting from the perturbed Hofstadter recursion and using the reduction identity from the previous subsection gives

$$Q(n) = Q(t_1(n)) + Q(t_1(n-1) + 1) + (-1)^n.$$

By the reconstruction formula $Q(m) = m + 1 - t_1(m + 1)$,

$$Q(t_1(n)) = t_1(n) + 1 - t_1(t_1(n) + 1), \quad Q(t_1(n-1) + 1) = t_1(n-1) + 2 - t_1(t_1(n-1) + 2).$$

Substitution yields

$$Q(n) = t_1(n) + 1 - t_1(t_1(n) + 1) + t_1(n-1) + 2 - t_1(t_1(n-1) + 2) + (-1)^n.$$

Finally, using $Q(n) = n + 1 - t_1(n + 1)$ and rearranging gives the stated recursion. $\square$

Theorem 4.4 (Equivalence with the clock recursion). *Let $t_1(n)$ be defined for $n \geq 2$ by $t_1(2) = 1$, $t_1(3) = 2$, and by the closed delayed recursion above for $n \geq 3$. Define $Q(n) = n + 1 - t_1(n + 1)$. Then $Q(1) = Q(2) = 1$, and $Q(n)$ satisfies the perturbed Hofstadter recursion*

$$Q(n) = Q(n - Q(n-1)) + Q(n - Q(n-2)) + (-1)^n.$$

Conversely, the clock process associated with any solution $Q(n)$ of the perturbed recursion satisfies this closed recursion.

Proof. The initial values give $Q(1) = 2 - t_1(2) = 1$ and $Q(2) = 3 - t_1(3) = 1$. By definition, $Q(n) = n + 1 - t_1(n + 1)$, while

$$n - Q(n-1) = t_1(n), \quad n - Q(n-2) = t_1(n-1) + 1.$$

Hence

$$\begin{aligned} Q(n - Q(n-1)) &= Q(t_1(n)) = t_1(n) + 1 - t_1(t_1(n) + 1), \\ Q(n - Q(n-2)) &= Q(t_1(n-1) + 1) = t_1(n-1) + 2 - t_1(t_1(n-1) + 2). \end{aligned}$$

Adding these two expressions and using the closed clock recursion gives

$$Q(n - Q(n-1)) + Q(n - Q(n-2)) + (-1)^n = n + 1 - t_1(n + 1) = Q(n).$$

The converse is exactly the derivation of the closed clock recursion from the original recurrence. $\square$

Thus the perturbed Hofstadter recursion may be reformulated equivalently as a closed delayed recursion for the auxiliary process $t_1(n)$.

4.6 Exact clock reduction formula

We now arrive at the central structural result of the paper. The renormalized defects $R(n) := Q(2n) - 2Q(n)$ admit an exact representation in terms of the auxiliary clock process $t_1(n)$.

Theorem 4.5 (Exact clock reduction formula). *For all $n \geq 1$,*

$$R(n) = Q(2n) - 2Q(n) = 2t_1(n + 1) - t_1(2n + 1) - 1.$$

Proof. By definition of the clock process, $t_1(m+1) = m+1-Q(m)$, hence $Q(m) = m+1-t_1(m+1)$. Applying this with $m = 2n$ and $m = n$ gives

$$Q(2n) = 2n + 1 - t_1(2n + 1), \quad Q(n) = n + 1 - t_1(n + 1).$$

Therefore

$$\begin{aligned} R(n) &= Q(2n) - 2Q(n) \\ &= (2n + 1 - t_1(2n + 1)) - 2(n + 1 - t_1(n + 1)) \\ &= 2t_1(n + 1) - t_1(2n + 1) - 1. \end{aligned}$$

□

Thus the renormalized defects are determined entirely by the auxiliary process $t_1(n)$. The following subsections investigate the consequences of this reduction for the large-scale scaling behavior of the recursion.

4.7 Parity structure of the clock dynamics

The clock formulation also reveals a simple parity structure underlying the recursive dynamics.

Proposition 4.6 (Parity classification of recursive arguments). *For every $n \geq 3$, the two recursive arguments $n - Q(n - 1)$ and $n - Q(n - 2)$ have the same parity:*

$$n - Q(n - 1) \equiv n - Q(n - 2) \equiv n - 1 \pmod{2}.$$

Equivalently, both recursive evaluations in the defining recursion occur at indices of parity opposite to n .

Proof. By Lemma 2.1, all values $Q(m)$ are odd. Hence $n - Q(n - 1) \equiv n - 1 \pmod{2}$, and similarly $n - Q(n - 2) \equiv n - 1 \pmod{2}$. □

Corollary 4.7 (Parity of the clock and defect sequences). *For all $n \geq 2$, $t_1(n) \equiv n - 1 \pmod{2}$. Moreover, the renormalized defects are always odd: $R(n) \equiv 1 \pmod{2}$.*

Proof. Since $t_1(n) = n - Q(n - 1)$ and all values $Q(n - 1)$ are odd by Lemma 2.1, we have $t_1(n) \equiv n - 1 \pmod{2}$. Similarly, both $Q(2n)$ and $Q(n)$ are odd, hence $R(n) = Q(2n) - 2Q(n)$ is odd. □

This parity separation is visible in the rescaled defect plots.

4.8 Consequences for dyadic scaling

The reduction formula $R(n) = 2t_1(n+1) - t_1(2n+1) - 1$ links the defect sequence across neighboring dyadic scales. In particular,

$$R(2n) = 2t_1(2n + 1) - t_1(4n + 1) - 1,$$

so successive dyadic levels are recursively coupled through the same auxiliary process.

5 Numerical Evidence for Logarithmic-Scale Organization

5.1 Dyadic fluctuation geometry

The preceding sections revealed that the centered fluctuations $E(n) = Q(n) - n/2$ and the renormalized defects $R(n) = Q(2n) - 2Q(n)$ exhibit reproducible behavior under dyadic rescaling.

To study this structure, consider the normalized fluctuation profiles on the dyadic grid. For $0 \leq j < 2^k$, set $x = j/2^k$ and define

$$E_k(x) := Q(2^k + j) - \frac{1}{2}(2^k + j).$$

Thus $E_k(x)$ records the centered fluctuations inside the dyadic block $[2^k, 2^{k+1})$ after rescaling the horizontal coordinate to unit length.

The dyadic collapse plots introduced earlier in Figure 7 show substantial alignment across scales together with visible parity separation. The renormalized defects provide an even clearer view of this behavior: since $R(n) = E(2n) - 2E(n)$, they measure precisely the failure of exact dyadic self-similarity of the centered fluctuations.

Moreover, the exact reduction formula $R(n) = 2t_1(n+1) - t_1(2n+1) - 1$ shows that the defect geometry is generated recursively through interactions between neighboring dyadic clock states.

5.2 Logarithmic oscillations and spectral evidence

Since the recursion repeatedly couples scales n and $2n$, it is natural to examine the fluctuations on the logarithmic scale $x = \log_2 n$. On this scale, dyadic rescaling becomes translation, since $\log_2(2n) = \log_2 n + 1$. Numerically, both the centered fluctuations $E(n) = Q(n) - n/2$ and the renormalized defects $R(n) = Q(2n) - 2Q(n)$ display structured oscillatory behavior when plotted against $\log_2 n$.

Figure 8 shows the dyadically normalized defect sequence. After normalization within successive dyadic blocks, the fluctuation profiles exhibit visible alignment across logarithmic scales.

Such alignment is reminiscent of phenomena associated with discrete scale invariance [11], but no limiting log-periodic law is proved here. A more quantitative Fourier analysis of these dyadic defect profiles is given in Subsection 6.2.

6 Scaling Heuristics and Asymptotic Interpretation

The numerical and structural results developed in the previous sections suggest persistent dyadic organization in the fluctuation structure of the recursion. After removal of the dominant linear growth term $Q(n) \sim n/2$, the remaining fluctuations exhibit persistent organization across multiple dyadic scales.

The centered fluctuations $E(n) = Q(n) - n/2$ and the renormalized defects $R(n) = Q(2n) - 2Q(n)$ therefore provide a natural framework for investigating the finer recursive dynamics hidden beneath the coarse asymptotic behavior.

6.1 Log-periodic evidence

The numerical data indicate that the fluctuations possess persistent oscillatory structure on the logarithmic scale $x = \log_2 n$. In particular, the dyadically rescaled fluctuation profiles display

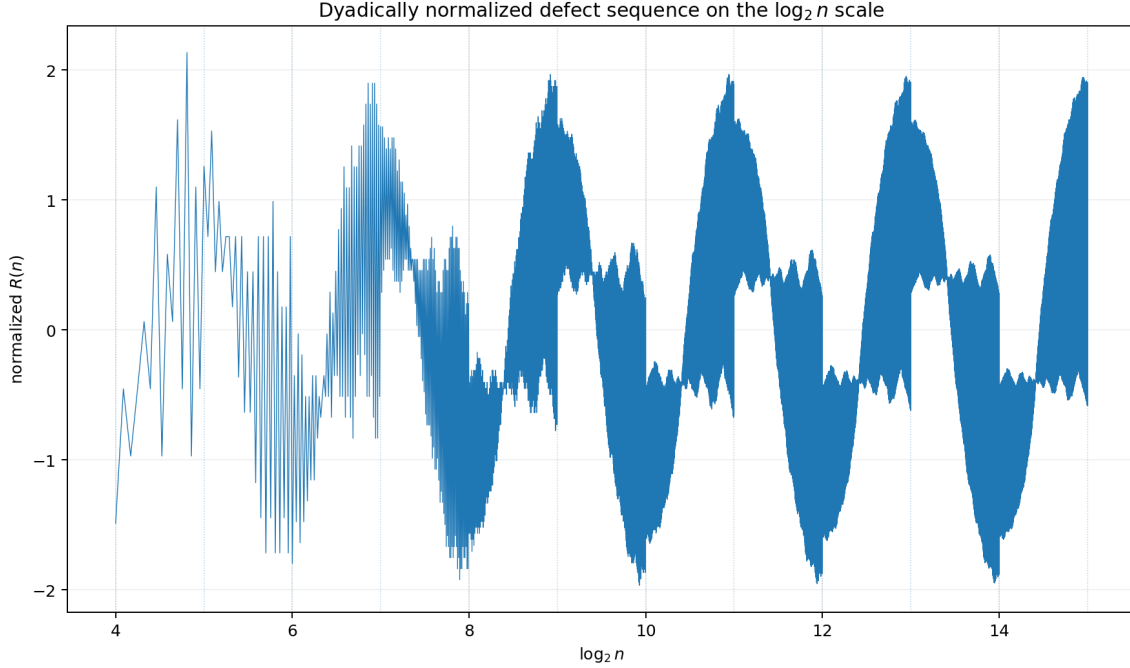


Figure 8: Dyadically normalized defect sequence plotted against $\log_2 n$. After normalization within dyadic blocks, the fluctuation profiles exhibit consistent alignment across logarithmic scales, suggesting possible logarithmic-scale organization.

substantial alignment across scales together with organized oscillatory envelopes. The normalized defect sequence likewise exhibits coherent modulation across successive dyadic scales.

Such behavior is reminiscent of phenomena associated with discrete scale invariance [11], where scaling symmetry occurs only under preferred dyadic rescalings. The persistence of these structures across all computed scales motivates further investigation.

6.2 Fourier evidence

Additional numerical evidence for recursive scaling behavior is provided by the Fourier spectra of the dyadically rescaled defect profiles. We describe the normalization and numerical protocol explicitly.

For each dyadic block

$$B_k = [2^k, 2^{k+1}), \quad 12 \leq k \leq 20,$$

we considered the block profile

$$R_k(j) := R(2^k + j), \quad 0 \leq j < 2^k.$$

The sequence $Q(n)$ was generated by direct exact integer recursion up to 2^{22} , which is sufficient for all values $R(n) = Q(2n) - 2Q(n)$ with $n < 2^{21}$, and hence for all blocks B_k with $k \leq 20$.

Before taking Fourier transforms, each block was standardized by subtracting its empirical mean

and dividing by its empirical standard deviation:

$$\tilde{R}_k(j) = \frac{R_k(j) - \mu_k}{\sigma_k}, \quad \mu_k = \frac{1}{2^k} \sum_{j=0}^{2^k-1} R_k(j),$$

and

$$\sigma_k^2 = \frac{1}{2^k} \sum_{j=0}^{2^k-1} (R_k(j) - \mu_k)^2.$$

No windowing was applied. Thus the reported spectra are spectra of the raw dyadic block profiles after blockwise centering and normalization. The discrete Fourier coefficients were computed as

$$\hat{R}_k(\ell) = \frac{1}{2^k} \sum_{j=0}^{2^k-1} \tilde{R}_k(j) \exp\left(-2\pi i \ell j / 2^k\right), \quad 0 \leq \ell < 2^k,$$

with power spectrum

$$P_k(\ell) = |\hat{R}_k(\ell)|^2.$$

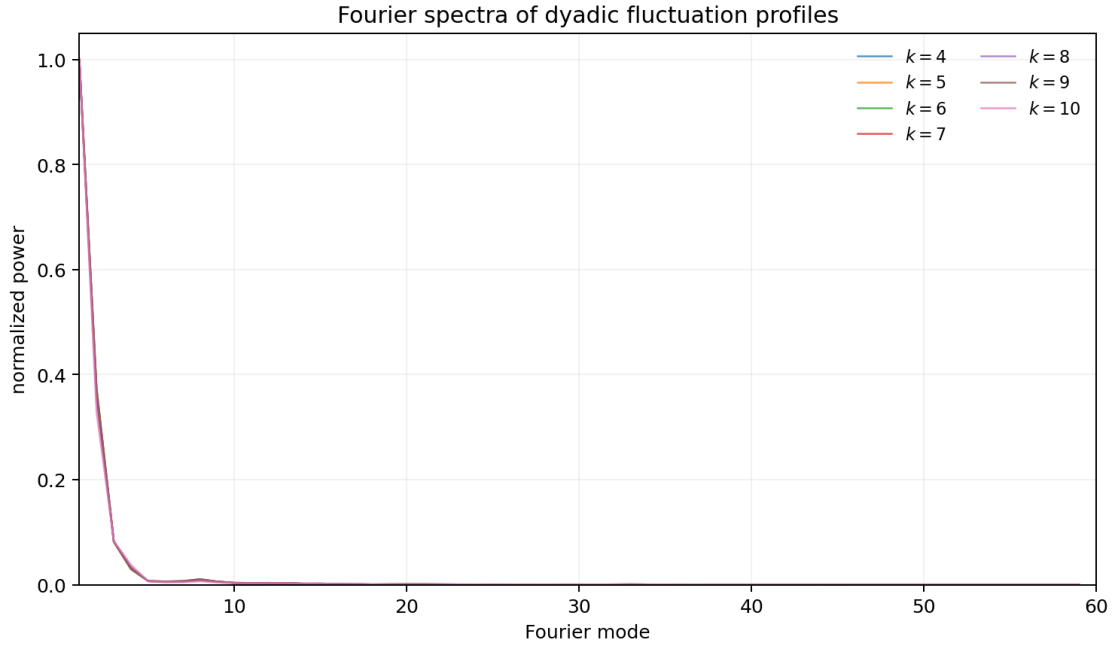


Figure 9: Discrete Fourier spectra of the dyadically rescaled fluctuation profiles. Several low-frequency modes remain visible across successive dyadic scales.

Since the profiles are centered, the zero mode is numerically negligible. For a fixed cutoff L , we define the low-frequency power fraction by

$$\rho_k(L) = \frac{\sum_{\ell=1}^L P_k(\ell) + \sum_{\ell=2^k-L}^{2^k-1} P_k(\ell)}{\sum_{\ell=1}^{2^k-1} P_k(\ell)}.$$

The second sum accounts for the conjugate negative-frequency modes of the real-valued profile.

The results show a persistent concentration of power in the first few Fourier modes. Table 1 gives the percentage of nonzero Fourier power contained in the first L positive and negative frequency modes for $L = 4, 8, 16, 32$.

k	2^k	$\rho_k(4)$	$\rho_k(8)$	$\rho_k(16)$	$\rho_k(32)$
12	4096	36.72%	38.07%	38.89%	39.30%
13	8192	36.32%	38.22%	39.04%	39.50%
14	16384	37.06%	38.40%	39.22%	39.64%
15	32768	36.54%	38.47%	39.28%	39.74%
16	65536	37.25%	38.59%	39.42%	39.86%
17	131072	36.72%	38.65%	39.44%	39.91%
18	262144	37.36%	38.73%	39.57%	40.00%
19	524288	36.88%	38.77%	39.57%	40.04%
20	1048576	37.43%	38.83%	39.68%	40.12%

Table 1: Low-frequency Fourier power fractions for the standardized dyadic defect profiles $\tilde{R}_k$. The entry $\rho_k(L)$ is the percentage of nonzero Fourier power contained in the first L positive and negative frequency modes. No windowing was applied.

The values in Table 1 are stable over the range $12 \leq k \leq 20$. For instance, the first eight positive and negative modes account for approximately 38% of the nonzero Fourier power throughout the computed range, while the first thirty-two modes account for about 40%. This stability supports the visual impression from Figure 9 that the dyadic defect profiles contain a persistent low-frequency component, rather than being dominated by scale-independent high-frequency variation.

Because the defect sequence has a strong parity structure, we also tested whether the observed low-frequency concentration is merely an artifact of even-odd separation. Two additional checks were performed. First, we subtracted separate empirical means on the even and odd sites of each block, i.e. on $j \equiv 0 \pmod{2}$ and $j \equiv 1 \pmod{2}$, before recomputing the Fourier spectra. Second, we computed spectra separately for the parity subprofiles

$$R_k^{(a)}(m) = R(2^k + 2m + a), \quad a \in \{0, 1\}, \quad 0 \leq m < 2^{k-1}.$$

The corresponding low-frequency fractions are shown in Table 2.

Figure 10 complements the parity robustness data in Table 2. For representative dyadic blocks, it shows the standardized parity subprofiles separately after rescaling the horizontal coordinate to unit length. More precisely, for $a \in \{0, 1\}$ each parity subprofile is standardized by

$$\tilde{R}_k^{(a)}(m) = \frac{R_k^{(a)}(m) - \mu_k^{(a)}}{\sigma_k^{(a)}}.$$

The visual alignment across scales indicates that the low-frequency structure persists within each parity class and is therefore not solely an artifact of alternating even-odd offsets.

Taken together, the power-fraction data and the parity-separated profiles suggest that the low-frequency spectral concentration is not explained solely by an even-odd offset. Rather, both parity classes appear to carry coherent large-scale structure across dyadic scales. The much larger low-frequency fractions for the separate parity subprofiles reflect the fact that restricting to one parity class removes the dominant alternating component and exposes a smoother large-scale profile.

k	$L = 8$				$L = 16$			
	all	p-dem.	even	odd	all	p-dem.	even	odd
12	38.07%	45.77%	95.66%	91.81%	38.89%	46.76%	97.76%	95.96%
14	38.40%	46.10%	95.72%	90.32%	39.22%	47.09%	97.77%	96.24%
16	38.59%	46.32%	95.73%	88.02%	39.42%	47.32%	97.75%	96.59%
18	38.73%	46.49%	95.73%	87.69%	39.57%	47.50%	97.75%	95.96%
20	38.83%	46.63%	95.72%	88.28%	39.68%	47.65%	97.75%	95.96%

Table 2: Parity robustness checks for the Fourier spectra of the dyadic defect profiles. The column “p-dem.” denotes the parity-demeaned profile, obtained by subtracting separate empirical means on the even and odd sites before standardization. The columns “even” and “odd” compute spectra on the two parity subprofiles separately. The persistence of low-frequency concentration after these operations indicates that the spectral structure is not solely caused by an even–odd mean offset.

These computations should be interpreted cautiously. They do not prove the existence of a limiting spectrum, a log-periodic fluctuation law, or a genuine discrete-scale-invariant limit. They do, however, provide reproducible numerical evidence that a substantial portion of the dyadic defect structure is organized at low Fourier modes and persists across the computed dyadic scales.

6.3 Heuristic dyadic scaling picture

Since $Q(n) \sim n/2$, the recursive arguments $n - Q(n - 1)$ and $n - Q(n - 2)$ typically occur near scale $n/2$. The defects $R(n) = Q(2n) - 2Q(n)$ therefore measure deviations from exact dyadic scaling. The exact reduction formula

$$R(n) = 2t_1(n + 1) - t_1(2n + 1) - 1$$

shows that these deviations are recursively coupled across neighboring dyadic scales. Although the present picture remains partly heuristic, the observed scaling behavior appears stable across the computed ranges.

6.4 Relation to existing meta-Fibonacci theories

A major recent advance was obtained by Cloître [2], who proved that the perturbed Hofstadter recursion is globally well-defined and satisfies

$$Q(n) = \frac{n}{2} + O\left(\frac{n}{\sqrt{\log n}}\right).$$

This establishes stable linear growth with asymptotic slope $1/2$, while still permitting substantial oscillatory fluctuation behavior.

The perturbed recursion also shares several qualitative features with classical meta-Fibonacci systems such as the Conolly and Tanny sequences [3, 12], including slow growth, structured frequency laws, and strong binary organization. There are, however, several important differences. The alternating perturbation $(-1)^n$ appears to generate substantially richer fluctuation geometry, including dyadic rescaling collapse, logarithmic oscillations, and persistent renormalized defect dynamics.

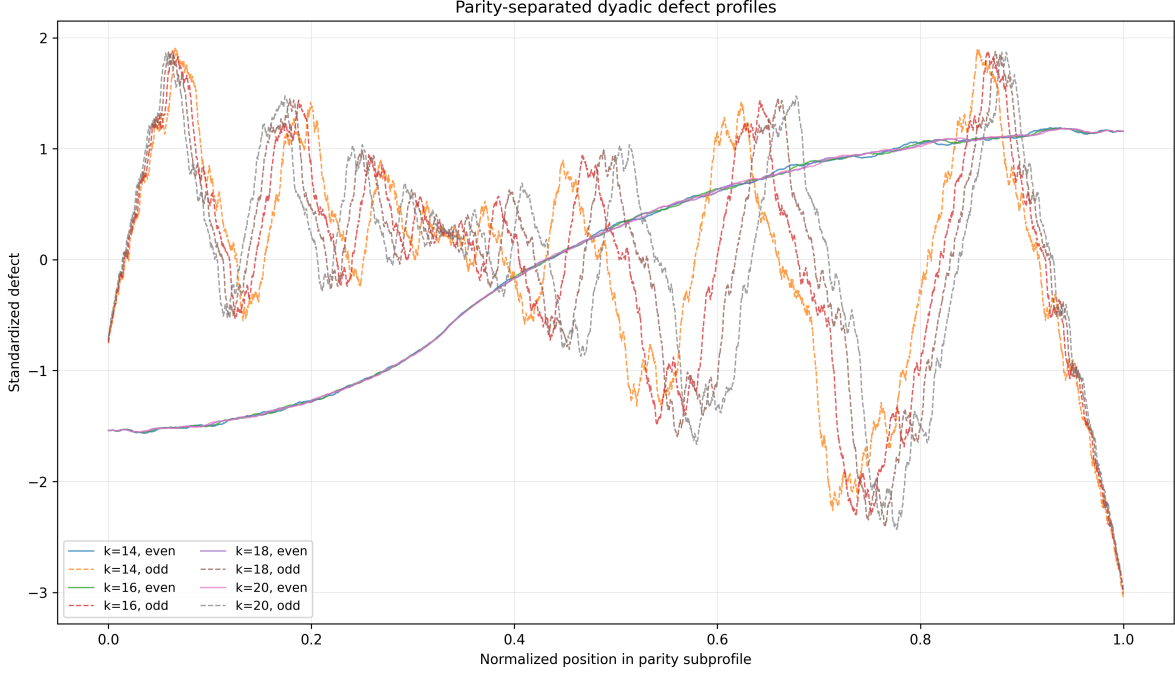


Figure 10: Parity-separated dyadic defect profiles for representative dyadic blocks $k \in \{14, 16, 18, 20\}$. The even and odd subprofiles are centered and normalized separately, and the horizontal coordinate is rescaled to the unit interval. Within each parity class, the profiles exhibit visible large-scale alignment across dyadic scales, supporting the interpretation that the low-frequency concentration seen in the Fourier spectra is not caused solely by a global even–odd alternation.

In particular, the exact identity

$$R(n) = 2t_1(n+1) - t_1(2n+1) - 1$$

has no known analogue in classical Hofstadter-type systems and directly connects recursive orbit timing with large-scale fluctuation geometry.

More broadly, the observed structures suggest that the perturbed recursion occupies an intermediate position between classical meta-Fibonacci systems and recursive scaling processes studied in analytic combinatorics and discrete dynamical systems. Together, Cloître’s asymptotic theorem and the defect framework developed here provide complementary information about both the coarse growth and finer fluctuation behavior of the recursion.

7 Open Problems and Future Directions

7.1 Boundedness and growth of $R(n)$

One of the central open problems concerns the asymptotic behavior of the renormalized defect sequence $R(n) = Q(2n) - 2Q(n)$. Cloître's estimate

$$Q(n) = \frac{n}{2} + O\left(\frac{n}{\sqrt{\log n}}\right)$$

implies $R(n) = O(n/\sqrt{\log n})$, but the numerical data suggest substantially smaller growth. Natural possibilities include bounded defects, $R(n) = O(1)$, polylogarithmic growth, or more general subpower behavior.

The clock reduction formula

$$R(n) = 2t_1(n+1) - t_1(2n+1) - 1$$

indicates that the asymptotic behavior of $R(n)$ is closely tied to the long-term dynamics of the clock process $t_1(n)$. A rigorous fluctuation theory for the perturbed recursion would therefore require substantially finer control of the recursive orbit structure.

7.2 Existence of genuine log-periodic structure

The data indicate that the fluctuations possess logarithmically organized oscillatory structure on the scale $x = \log_2 n$. In particular, the centered fluctuations $E(n) = Q(n) - n/2$ and the renormalized defects $R(n) = Q(2n) - 2Q(n)$ appear to exhibit persistent modulation across dyadic scales.

A central open question is whether these oscillations correspond to genuine asymptotic logarithmic periodicity or merely to large finite-scale effects. More precisely, one may ask whether there exist nonconstant periodic functions, say $\Phi(x+1) = \Phi(x)$ and $\Psi(x+1) = \Psi(x)$, such that

$$E(n) \approx \Phi(\log_2 n) \quad \text{or} \quad R(n) \approx \Psi(\log_2 n)$$

in an appropriate asymptotic sense.

The observed dyadic alignment and spectral coherence are reminiscent of discrete-scale-invariant behavior, but no rigorous mechanism is currently known.

7.3 Higher-order dyadic defect operators

The defects $R(n) = Q(2n) - 2Q(n)$ capture only the first level of dyadic scaling corrections. A natural next step is to investigate higher-order renormalized quantities such as $R_2(n) = R(2n) - 2R(n)$, or more generally the iterated renormalization operator

$$(\mathcal{R}f)(n) = f(2n) - 2f(n).$$

The numerical data suggest that repeated renormalization may reveal progressively finer fluctuation structures hidden beneath the dominant scaling law. It would therefore be interesting to determine whether higher-order defects remain bounded, converge under rescaling, or generate stable recursive scaling hierarchies.

7.4 Extensions to other perturbations

The present work studies the alternating perturbation $(-1)^n$, which appears closely synchronized with the dyadic structure of the recursion. A natural problem is to investigate more general perturbations of the form

$$Q_f(n) = Q_f(n - Q_f(n - 1)) + Q_f(n - Q_f(n - 2)) + f(n),$$

where $f(n)$ is a deterministic forcing sequence. Particularly interesting cases include periodic, automatic, digital, or pseudorandom perturbations. An important question is which perturbations preserve stable asymptotic growth, explicit frequency laws, and coherent dyadic fluctuation structures.

7.5 Toward a general theory of dyadic nested recursions

The numerical and structural results obtained in this paper suggest that certain nested recursions may exhibit stable large-scale scaling behavior together with nontrivial fluctuation patterns. Several features are now visible: proved asymptotic linear growth from Cloître’s work, proved explicit frequency laws, dyadic orbit organization, numerical logarithmic oscillations, and apparent organization across dyadic scales.

These phenomena resemble structures appearing in divide-and-conquer systems, digital combinatorics, and recursive dynamical systems. The perturbed Hofstadter recursion could therefore provide a useful experimental model for studying recursive dyadic scaling mechanisms in self-referential systems. A remaining open problem is to understand whether the same rank-lifting mechanism persists for other initial seeds or for other periodic perturbations.

At present, however, basic questions concerning limiting distributions of the defect profiles, spectral structure, and large-scale fluctuation dynamics remain open. The perturbed Hofstadter recursion therefore provides an unusual example of a self-referential integer recursion combining explicit frequency laws with stable asymptotic growth.

8 Acknowledgments

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A Cloître Interleaving and Dyadic Hitting-Time Identities

This appendix proves the dyadic hitting-time identity

$$T(2^k) = 2^{k+2} - k - 2,$$

used in Section 3. The proof uses Cloître’s binary interleaving construction for the monotone subsequences $A(m) = (Q(2m - 1) + 1)/2$ and $B(m) = (Q(2m) + 1)/2$.

A.1 The interleaving operation

We recall the interleaving operation on binary words.

Definition A.1 (Interleaving). Let X and Y be finite binary words, and let $b \in \{0, 1\}$. The word $\text{Interleave}(X, Y, b)$ is produced by a two-tape machine as follows. The machine has state b . In state 0 it reads the next symbol of X , and in state 1 it reads the next symbol of Y . The symbol read is emitted and becomes the new state. If the prescribed tape is empty and the other tape is not empty, the machine reads from the other tape.

If $Z = \text{Interleave}(X, Y, b)$, we denote by i_t and j_t the numbers of symbols consumed from X and Y , respectively, after t output symbols have been produced.

For a binary word W , define

$$Z_W(t) := \#\{0 \leq n < t : W(n) = 0\}, \quad O_W(t) := \#\{0 \leq n < t : W(n) = 1\},$$

and $h_W(t) := Z_W(t) - O_W(t)$.

Lemma A.2 (Height additivity). *If $Z = \text{Interleave}(X, Y, b)$, then for every t ,*

$$h_Z(t) = h_X(i_t) + h_Y(j_t).$$

Proof. The prefix $Z[0 : t)$ consists of the two prefixes $X[0 : i_t)$ and $Y[0 : j_t)$, interleaved. Interleaving changes the order but not the number of zeroes or ones. Hence the zero-counts and one-counts are additive, and so are the heights. $\square$

Definition A.3 (Extractions). For a binary word W , let $E_0(W)$ be the word consisting of the first bit of W together with every bit immediately preceded by a 0. Let $E_1(W)$ be the word consisting of every bit immediately preceded by a 1.

Lemma A.4 (Interleaving extraction). *If $W = \text{Interleave}(X, Y, 0)$, then*

$$E_0(W) = X, \quad E_1(W) = Y.$$

Proof. In the interleaving machine with initial state 0, the first emitted symbol is read from the X -tape. After that, the current state is always the previously emitted bit. Hence every symbol emitted immediately after a 0 is read from the X -tape, and every symbol emitted immediately after a 1 is read from the Y -tape. Therefore the extraction words recover exactly the two input tapes. $\square$

We shall also use the reverse-complement operation. If $W = w_0 w_1 \cdots w_{n-1}$, define

$$\text{rc}(W) := (1 - w_{n-1})(1 - w_{n-2}) \cdots (1 - w_0).$$

A.2 Cloître's arch words

Cloître's construction decomposes the step words of A and B into positive and negative arch words P_r and N_r . We use the notation $a_0 = 3$, $a_{r+1} = 4a_r - 1$, so that

$$a_r = \frac{2 \cdot 4^{r+1} + 1}{3}.$$

Define $u_r = 2a_r - r - 2$ and $v_r = u_r + 2a_r = 4a_r - r - 2$. The positive arch of level r is the interval $[u_r, v_r]$, and the negative arch is $[v_r, u_{r+1}]$.

The corresponding arch words are

$$P_r = (\Delta A(u_r), \Delta A(u_r + 1), \dots, \Delta A(v_r - 1)), \quad N_r = (\Delta B(v_r), \Delta B(v_r + 1), \dots, \Delta B(u_{r+1} - 1)).$$

The following two interleaving laws are due to Cloître [2]. In Cloître's notation they arise from the arch decomposition of the parity subsequences and are unconditional after the proof of the global arch skeleton and the 1-Lipschitz property.

$$N_r = \text{Interleave}(P_r[2:], P_r[: -1], 1), \quad (\text{L1})$$

and

$$P_{r+1} = \text{Interleave}([0, 0] \circ N_r \circ [1], [0] \circ N_r, 0). \quad (\text{L2})$$

The initial positive word is $P_0 = 001011$. For later use set $q_r := (4^{r+1} - 1)/3$. Then

$$a_r = 2q_r + 1, \quad |P_r| = 2a_r = 4q_r + 2, \quad |N_r| = 8q_r + 1.$$

A.3 Quarter-crossing times

We define the positive quarter-crossing times in P_r by

$$\alpha_r := \min\{t : O_{P_r}(t) = q_r\}, \quad \beta_r := \min\{t : Z_{P_r}(t) = q_r\}.$$

Similarly, set $q_r^- := 2q_r$. The negative quarter-crossing times in N_r are

$$\alpha_r^- := \min\{t : Z_{N_r}(t) = q_r^- - 1\} = \min\{t : Z_{N_r}(t) = 2q_r - 1\},$$

and

$$\beta_r^- := \min\{t : O_{N_r}(t) = q_r^-\} = \min\{t : O_{N_r}(t) = 2q_r\}.$$

The key quarter-crossing identities are

$$\alpha_r + \beta_r = 4q_r, \quad (\text{Q+})$$

and

$$\alpha_r^- + \beta_r^- = 4q_r^- - 2 = 8q_r - 2. \quad (\text{Q-})$$

Equivalently, these are height identities.

Lemma A.5 (Quarter-crossings as height equalities). *The identity (Q+) is equivalent to $h_{P_r}(\alpha_r) = h_{P_r}(\beta_r)$, and (Q-) is equivalent to $h_{N_r}(\alpha_r^-) = h_{N_r}(\beta_r^-)$.*

Proof. At $t = \alpha_r$, we have $O_{P_r}(\alpha_r) = q_r$, hence

$$h_{P_r}(\alpha_r) = (\alpha_r - q_r) - q_r = \alpha_r - 2q_r.$$

At $t = \beta_r$, we have $Z_{P_r}(\beta_r) = q_r$, hence

$$h_{P_r}(\beta_r) = q_r - (\beta_r - q_r) = 2q_r - \beta_r.$$

Thus $h_{P_r}(\alpha_r) = h_{P_r}(\beta_r)$ is equivalent to $\alpha_r + \beta_r = 4q_r$.

The negative case is analogous. At $t = \alpha_r^-$, we have $Z_{N_r}(\alpha_r^-) = 2q_r - 1$, so

$$h_{N_r}(\alpha_r^-) = (2q_r - 1) - (\alpha_r^- - (2q_r - 1)) = 4q_r - 2 - \alpha_r^-.$$

At $t = \beta_r^-$, we have $O_{N_r}(\beta_r^-) = 2q_r$, so

$$h_{N_r}(\beta_r^-) = (\beta_r^- - 2q_r) - 2q_r = \beta_r^- - 4q_r.$$

Equality of these heights is equivalent to $\alpha_r^- + \beta_r^- = 8q_r - 2$. □

A.4 Reverse-complement symmetry

We shall use a reverse-complement symmetry of the arch words. This symmetry is not an additional assumption; it follows from Cloître's interleaving laws for P_r and N_r . We give the proof here for completeness.

Lemma A.6 (Truncated Law-1 reverse-complement symmetry). *Assume that $P_r = \text{rc}(P_r)$. Then the truncated negative word $N_r[0 : |N_r| - 1]$ is reverse-complementary.*

Proof. By Law 1,

$$N_r = \text{Interleave}(P_r[2:], P_r[: -1], 1).$$

The final symbol of N_r is the terminal boundary symbol arising from the longer second tape $P_r[: -1]$. Removing this final boundary symbol gives

$$N_r[0 : |N_r| - 1] = \text{Interleave}(P_r[2:], P_r[: -2], 1).$$

Since $P_r = \text{rc}(P_r)$, we have $P_r[: -2] = \text{rc}(P_r[2:])$. Thus the two tapes in the truncated Law-1 interleaving are reverse-complements of one another.

The interleaving machine is reversible under reverse-complementation: reading the output backwards and complementing all symbols exchanges the two states 0 and 1, and therefore exchanges the two tapes. Since the two truncated tapes are reverse-complements, the reversed-complemented interleaving run uses exactly the same tape data in the opposite order. Hence $N_r[0 : |N_r| - 1]$ is reverse-complementary. □

Lemma A.7 (Law-2 reverse-complement propagation). *Assume that $N_r[0 : |N_r| - 1]$ is reverse-complementary. Then $P_{r+1} = \text{rc}(P_{r+1})$.*

Proof. By Law 2,

$$P_{r+1} = \text{Interleave}([0, 0] \circ N_r \circ [1], [0] \circ N_r, 0).$$

Write $N_r = N_r^\circ \circ \eta$, where $N_r^\circ := N_r[0 : |N_r| - 1]$ and η is the final boundary symbol of N_r . By assumption, $N_r^\circ = \text{rc}(N_r^\circ)$.

After separating the boundary symbol η , the two Law-2 tapes $[0, 0] \circ N_r \circ [1]$ and $[0] \circ N_r$ are paired by reverse-complementation, with the initial padding of one tape matching the terminal padding of the other. Reversing the output and complementing all bits therefore exchanges the two tapes and reconstructs the same interleaving run in the opposite direction. Hence $P_{r+1} = \text{rc}(P_{r+1})$. □

Lemma A.8 (Positive reverse-complement symmetry). *For every $r \geq 0$, $P_r = \text{rc}(P_r)$.*

Proof. The claim holds for $r = 0$, since $P_0 = 001011 = \text{rc}(001011)$. Assume now that $P_r = \text{rc}(P_r)$. By Lemma A.6, $N_r[0 : |N_r| - 1]$ is reverse-complementary. Applying Lemma A.7 gives $P_{r+1} = \text{rc}(P_{r+1})$. The result follows by induction. $\square$

Lemma A.9 (Negative truncated reverse-complement symmetry). *Let $L_r := |N_r| = 8q_r + 1$. Then the truncated word $N_r[0 : L_r - 1]$ is reverse-complementary. Equivalently, for $0 \leq t \leq L_r - 2$,*

$$N_r(t) = 1 - N_r(L_r - 2 - t).$$

Proof. By Lemma A.8, $P_r = \text{rc}(P_r)$. Therefore Lemma A.6 implies that

$$N_r[0 : |N_r| - 1] = N_r[0 : L_r - 1]$$

is reverse-complementary. $\square$

A.5 The Law-1 step

We first prove that the positive quarter identity implies the negative quarter identity.

Lemma A.10 (Law-1 parking rule). *In the Law-1 interleaving $N_r = \text{Interleave}(P_r[2:], P_r[: -1], 1)$, let $W = N_r[0 : |N_r| - 1]$. If $a = \min\{t : Z_W(t) = 2q_r - 1\}$, then the antipodal position $b := |W| - 2 - a$ is the first position at which the one-count reaches $2q_r$, that is,*

$$b = \min\{t : O_W(t) = 2q_r\}.$$

Proof. By Lemma A.9, the truncated Law-1 word $W = N_r[0 : |N_r| - 1]$ is reverse-complementary. Hence the antipodal map $t \mapsto |W| - 2 - t$ exchanges zero-count first passages on the left side of W with one-count first passages on the right side.

At the time $a = \min\{t : Z_W(t) = 2q_r - 1\}$, the prefix $W[0 : a]$ has just accumulated the required number of zeroes. Under reverse-complementation, the complementary suffix corresponds to a prefix ending at $b = |W| - 2 - a$ with exactly $2q_r$ ones. Thus $O_W(b) = 2q_r$.

It remains to check minimality. If some $b' < b$ satisfied $O_W(b') = 2q_r$, then applying the antipodal reverse-complement map would produce a position $a' < a$ with $Z_W(a') = 2q_r - 1$, contradicting the defining minimality of a . Therefore $b = \min\{t : O_W(t) = 2q_r\}$. $\square$

Lemma A.11 (Law-1 quarter antipode). *Let $W = N_r[0 : |N_r| - 1]$. If $a = \alpha_r^-$, then*

$$\beta_r^- = |W| - 2 - a.$$

Proof. Set $|W| = 8q_r$. By Lemma A.9, the word W is reverse-complementary, and hence contains exactly $4q_r$ zeroes and $4q_r$ ones.

Let $a = \alpha_r^-$. By definition, $Z_W(a) = 2q_r - 1$. Since a is a first-passage time, the next relevant symbol is the zero which realizes this count. Define $b := |W| - 2 - a$. By Lemma A.10, this position is the first position at which the one-count reaches $2q_r$, so

$$b = \min\{t : O_W(t) = 2q_r\}.$$

Since $W = N_r[0 : |N_r| - 1]$ contains the relevant negative quarter crossing before the removed terminal boundary symbol, this is precisely $b = \beta_r^-$. Therefore $\beta_r^- = |W| - 2 - a$. $\square$

Proposition A.12 (Law-1 quarter step). *For every $r \geq 0$,*

$$\alpha_r + \beta_r = 4q_r \implies \alpha_r^- + \beta_r^- = 8q_r - 2.$$

Equivalently,

$$h_{P_r}(\alpha_r) = h_{P_r}(\beta_r) \implies h_{N_r}(\alpha_r^-) = h_{N_r}(\beta_r^-).$$

Proof. Let $W = N_r[0 : |N_r| - 1]$, so that $|W| = |N_r| - 1 = 8q_r$, and put $a := \alpha_r^-$. By Lemma A.11, $\beta_r^- = |W| - 2 - a$. Therefore

$$\alpha_r^- + \beta_r^- = a + (|W| - 2 - a) = |W| - 2 = 8q_r - 2.$$

By Lemma A.5, this is equivalent to $h_{N_r}(\alpha_r^-) = h_{N_r}(\beta_r^-)$. $\square$

A.6 The Law-2 step

We now prove the converse propagation from N_r to P_{r+1} . Let

$$P_{r+1} = \text{Interleave}(X, Y, 0), \quad X = [0, 0] \circ N_r \circ [1], \quad Y = [0] \circ N_r.$$

Let i_t, j_t denote the consumed tape lengths at time t . At the positive quarter times α_{r+1} and β_{r+1} , define the inner head coordinates

$$I_\alpha := i_{\alpha_{r+1}} - 2, \quad J_\alpha := j_{\alpha_{r+1}} - 1, \quad I_\beta := i_{\beta_{r+1}} - 2, \quad J_\beta := j_{\beta_{r+1}} - 1.$$

Lemma A.13 (Law-2 count pairings). *The following identities hold:*

$$O_{N_r}(I_\alpha) = \alpha_r, \quad O_{N_r}(J_\alpha) = \beta_r, \quad Z_{N_r}(I_\beta) = \alpha_r - 2, \quad Z_{N_r}(J_\beta) = \beta_r - 1.$$

Proof. At the time α_{r+1} , the word P_{r+1} has just reached $O_{P_{r+1}}(\alpha_{r+1}) = q_{r+1} = 4q_r + 1$. Immediately before this final one is emitted, the prefix contains $4q_r$ ones. Since X and Y have only zeroes in their initial padding, these $4q_r$ ones come from the two N_r -copies. Hence

$$O_{N_r}(I_\alpha) + O_{N_r}(J_\alpha) = 4q_r.$$

By Lemma A.4, $E_1(P_{r+1}) = Y = [0] \circ N_r$. Thus the Y -head is positioned so that $O_{N_r}(J_\alpha) = \beta_r$. Using $\alpha_r + \beta_r = 4q_r$, we obtain $O_{N_r}(I_\alpha) = \alpha_r$.

The zero-count identities are analogous. At the time β_{r+1} , the word P_{r+1} has just reached $Z_{P_{r+1}}(\beta_{r+1}) = q_{r+1} = 4q_r + 1$. Before the final zero is emitted, the prefix contains $4q_r$ zeroes. The two initial zeroes in $X = [0, 0] \circ N_r \circ [1]$ and the one initial zero in $Y = [0] \circ N_r$ account for three of these zeroes. Therefore

$$Z_{N_r}(I_\beta) + Z_{N_r}(J_\beta) = 4q_r - 3.$$

Again by Lemma A.4, $E_1(P_{r+1}) = Y = [0] \circ N_r$. Hence $Z_{N_r}(J_\beta) = \beta_r - 1$, and so

$$Z_{N_r}(I_\beta) = 4q_r - 3 - (\beta_r - 1) = 4q_r - \beta_r - 2 = \alpha_r - 2.$$

$\square$

Lemma A.14 (Quarter antipode identities). *With $L = |N_r| = 8q_r + 1$, one has*

$$I_\alpha + J_\beta = L - 2, \quad J_\alpha + I_\beta = L - 4.$$

Proof. The second identity follows directly from the head counts. At the α -quarter crossing, the Y -head has consumed $4q_r$ symbols including the initial padding zero, hence $J_\alpha = 4q_r - 1$. At the β -quarter crossing, the X -head has consumed $4q_r$ symbols including the two initial padding zeroes, hence $I_\beta = 4q_r - 2$. Thus

$$J_\alpha + I_\beta = (4q_r - 1) + (4q_r - 2) = 8q_r - 3 = L - 4.$$

For the first identity, use the reverse-complement symmetry of $N_r[0 : L - 1]$. The α -quarter X -head stands at a position I_α in N_r with $O_{N_r}(I_\alpha) = \alpha_r$. The antipodal position in the truncated reverse-complement word is $J^* := L - 2 - I_\alpha$. By reverse-complement symmetry, the zero-count before J^* is $Z_{N_r}(J^*) = \beta_r - 1$. The Y -head position J_β is characterized by the same zero-count plateau, $Z_{N_r}(J_\beta) = \beta_r - 1$, and by the same terminal zero event. Hence $J_\beta = J^*$, and therefore $I_\alpha + J_\beta = L - 2$. $\square$

Lemma A.15 (Inner quarter-height balance). *One has*

$$h_{N_r}(I_\alpha) + h_{N_r}(J_\alpha) = h_{N_r}(I_\beta) + h_{N_r}(J_\beta) + 2.$$

Proof. Using $J_\alpha + I_\beta = L - 4$ together with the count pairings of Lemma A.13, we get

$$h_{N_r}(J_\alpha) - h_{N_r}(I_\beta) = Z_{N_r}(J_\alpha) + O_{N_r}(I_\beta) - \alpha_r - \beta_r + 2.$$

The identity $J_\alpha + I_\beta = L - 4$ implies $Z_{N_r}(J_\alpha) + O_{N_r}(I_\beta) = 4q_r - 1$, and since $\alpha_r + \beta_r = 4q_r$, we obtain

$$h_{N_r}(J_\alpha) = h_{N_r}(I_\beta) + 1.$$

Similarly, using $I_\alpha + J_\beta = L - 2$, we obtain

$$h_{N_r}(I_\alpha) = h_{N_r}(J_\beta) + 1.$$

Adding the two identities gives the claim. $\square$

Proposition A.16 (Law-2 quarter step). *For every $r \geq 0$,*

$$h_{N_r}(\alpha_r^-) = h_{N_r}(\beta_r^-) \implies h_{P_{r+1}}(\alpha_{r+1}) = h_{P_{r+1}}(\beta_{r+1}).$$

Equivalently,

$$\alpha_r^- + \beta_r^- = 8q_r - 2 \implies \alpha_{r+1} + \beta_{r+1} = 4q_{r+1}.$$

Proof. By height additivity, $h_{P_{r+1}}(t) = h_X(i_t) + h_Y(j_t)$. At the α -quarter crossing, the final emitted symbol is a one. Accounting for the initial padding in X and Y , we obtain

$$h_{P_{r+1}}(\alpha_{r+1}) = 2 + h_{N_r}(I_\alpha) + h_{N_r}(J_\alpha).$$

At the β -quarter crossing, the final emitted symbol is a zero, and the padding contributes one additional unit. Thus

$$h_{P_{r+1}}(\beta_{r+1}) = 4 + h_{N_r}(I_\beta) + h_{N_r}(J_\beta).$$

The result follows immediately from Lemma A.15. $\square$

A.7 Proof of the quarter-crossing identities

We now prove (Q+) and (Q-) for all $r \geq 0$.

Theorem A.17 (Quarter-crossing identities). *For every $r \geq 0$,*

$$\alpha_r + \beta_r = 4q_r, \quad \alpha_r^- + \beta_r^- = 8q_r - 2.$$

Proof. For $r = 0$, we have $P_0 = 001011$ and $q_0 = 1$. Hence $\alpha_0 = 3$, $\beta_0 = 1$, and therefore $\alpha_0 + \beta_0 = 4 = 4q_0$. Thus the positive quarter identity holds at $r = 0$.

By Proposition A.12, the positive quarter identity at level r implies the negative quarter identity at the same level. By Proposition A.16, the negative quarter identity at level r implies the positive quarter identity at level $r + 1$. Hence

$$Q_r^+ \implies Q_r^- \implies Q_{r+1}^+$$

for all $r \geq 0$. Starting from Q_0^+ , both quarter-crossing identities follow for every r . $\square$

A.8 Dyadic hitting-time endpoints

We now derive the dyadic endpoint formula. Recall that $T(s) = \tau_A(s) + \tau_B(s)$.

Theorem A.18 (Dyadic hitting-time endpoints). *For every $k \geq 0$,*

$$T(2^k) = 2^{k+2} - k - 2.$$

Proof. The case $k = 0$ is immediate: since $A(1) = B(1) = 1$, we have $\tau_A(1) = \tau_B(1) = 1$, and hence $T(1) = 2 = 2^2 - 2$.

For $k \geq 1$, we distinguish even and odd dyadic exponents.

First let $2^k = 4^{r+1}$, so $k = 2r + 2$, and set $p = 4^{r+1}$. Then p lies inside the positive arch of level r , and $q := p - a_r = (4^{r+1} - 1)/3 = q_r$. On the positive arch,

$$A(u_r + t) = a_r + O_{P_r}(t), \quad B(u_r + t) = a_r + Z_{P_r}(t).$$

Therefore $\tau_A(p) = u_r + \alpha_r$ and $\tau_B(p) = u_r + \beta_r$, whence

$$T(p) = 2u_r + \alpha_r + \beta_r.$$

Using $u_r = 2a_r - r - 2$ and $\alpha_r + \beta_r = 4q_r = 4(p - a_r)$, we obtain

$$T(p) = 2(2a_r - r - 2) + 4(p - a_r) = 4p - 2r - 4.$$

Since $k = 2r + 2$, this gives

$$T(p) = 4p - k - 2 = 2^{k+2} - k - 2.$$

Now let $2^k = 2 \cdot 4^{r+1}$, so $k = 2r + 3$, and set $p = 2 \cdot 4^{r+1}$. This value lies in the negative arch following the positive arch of level r . Put $c := 2a_r$. Then $q := p - c = 2(4^{r+1} - 1)/3 = 2q_r = q_r^-$. On the negative arch,

$$B(v_r + t) = c + O_{N_r}(t), \quad A(v_r + t + 1) = c + Z_{N_r}(t) + 1.$$

Therefore $\tau_B(p) = v_r + \beta_r^-$ and $\tau_A(p) = v_r + 1 + \alpha_r^-$, whence

$$T(p) = 2v_r + 1 + \alpha_r^- + \beta_r^-.$$

Using $v_r = 4a_r - r - 2$ and $\alpha_r^- + \beta_r^- = 4q_r^- - 2 = 4(p - 2a_r) - 2$, we obtain

$$T(p) = 2(4a_r - r - 2) + 1 + 4(p - 2a_r) - 2 = 4p - 2r - 5.$$

Since $k = 2r + 3$, this gives

$$T(p) = 4p - k - 2 = 2^{k+2} - k - 2.$$

Thus in both cases $T(2^k) = 2^{k+2} - k - 2$. □

B Computational Methods and Numerical Data

This appendix summarizes the computational procedures, verification checks, and representative numerical data used throughout the paper. All computations were performed using exact integer arithmetic and direct recursive evaluation of $Q(1) = Q(2) = 1$ and

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n.$$

All numerical phenomena reported in the experimental sections were verified up to at least $N = 10^7$, unless a smaller range is explicitly stated for a particular figure or diagnostic.

The computations were carried out on a PC equipped with an Intel Core i9-14900K processor with 24 cores and 64 GB RAM. The implementation used exact integer arithmetic throughout; no floating-point arithmetic was used in the generation of the sequence itself. Floating-point arithmetic was used only for postprocessing tasks such as normalization, plotting, and Fourier transforms.

The code and data used to generate the numerical results are available at

<https://github.com/Mantovanelli1110/perturbed-hofstadter-q>.

The repository contains the scripts used to generate the figures, compute the Fourier diagnostics, and perform the numerical verification checks described in this appendix. An archived version with a persistent DOI, together with the corresponding commit hash and data checksums, will be provided after acceptance or upon request.

Representative numerical data

The first values of the sequence are

$$(Q(n))_{n \geq 1} = (1, 1, 1, 3, 3, 3, 5, 5, 5, 7, 5, 9, 7, 9, 7, 11, 9, 11, 11, 11, \dots).$$

Only odd values occur, consistent with the parity property proved in the main text.

The first values of the defect sequence $R(n) = Q(2n) - 2Q(n)$ are

$$(R(n))_{n \geq 1} = (-1, -1, -1, -1, -5, -1, -5, -1, -9, -1, -5, -1, \dots).$$

The corresponding initial frequency values are

m	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$F(m)$	3	3	4	3	3	5	4	3	3	4	6	5	3	3	4	3	3

The computations provide independent checks of the exact identities proved above and support the additional numerical phenomena discussed in the later sections.

Auxiliary identities used in the computations

Recall the centered fluctuation function $E(n) = Q(n) - n/2$. The defect sequence was computed as

$$R(n) = Q(2n) - 2Q(n) = E(2n) - 2E(n),$$

so that $R(n)$ measures deviations from exact dyadic scaling of the centered fluctuations.

For the frequency computations, only odd values occur, so values were parameterized as $2m - 1$. The frequency function $F(m) = \#\{n \geq 1 : Q(n) = 2m - 1\}$ was evaluated over all $1 \leq n \leq N$ and then restricted to values whose full occurrence set was contained in the computed range. The cumulative counting function was computed as $C(m) = \sum_{s \leq m} F(s)$.

Direct recursive generation

The sequence was generated sequentially. Since the recursive arguments at time n involve only values $Q(j)$ with $j < n$, once positivity of the recursive arguments is checked, the value $Q(n)$ can be computed directly from previously stored terms.

The following pseudocode describes the basic computation.

```

Input:  $N$ 
Output:  $Q(1), \dots, Q(N)$ 
 $Q(1) \leftarrow 1, \quad Q(2) \leftarrow 1$ 
for  $n = 3, \dots, N$  do
     $a \leftarrow n - Q(n - 1)$ 
     $b \leftarrow n - Q(n - 2)$ 
    if  $a \leq 0$  or  $b \leq 0$  then stop
     $Q(n) \leftarrow Q(a) + Q(b) + (-1)^n$ 
end for

```

This algorithm requires $O(N)$ time and $O(N)$ memory. The storage of the full sequence is convenient because the recursion uses nonlocal earlier values $Q(a)$ and $Q(b)$.

Figure data ranges

The figures in the paper were generated directly from the exact recursive datasets without smoothing or artificial interpolation. The following table summarizes the numerical ranges used for the main plots. The filenames listed here are the figure files used in the L^AT_EX source.

For the dyadic collapse plots, each dyadic block $[2^k, 2^{k+1})$ was normalized independently before overlaying the corresponding profiles. For the parity-separated profiles, the even and odd subprofiles were centered and normalized separately.

Fourier analysis

For the Fourier analysis, the sequence was generated by exact integer recursion up to 2^{22} , which is sufficient for all defect values $R(n) = Q(2n) - 2Q(n)$ with $n < 2^{21}$. We analyzed the dyadic defect blocks $B_k = [2^k, 2^{k+1})$ for $12 \leq k \leq 20$. On each block, the profile $R_k(j) = R(2^k + j)$, $0 \leq j < 2^k$, was centered and divided by its empirical standard deviation before applying the discrete Fourier transform.

Figure file	Quantity plotted	Range
firstvalues_4096.png	$Q(n)$	$1 \leq n \leq 4096$
perturbed_hofstadter_safety_margin_clean.png	$S(n) = n - \max\{Q(n-1), Q(n-2)\}$	$3 \leq n \leq 4096$
perturbed_hofstadter_first_differences_clean.png	$D(n) = Q(n+1) - Q(n)$	$1 \leq n < 4096$
frequency_distribution.png	$F(m)$ and $M(m) = \max_{1 \leq j \leq m} F(j)$	$1 \leq m \leq 4000$
dyadic_frequency_blocks_B6_B9.png	$F(m)$ on dyadic blocks	$6 \leq k \leq 9$
normalized_peak_locations.png	normalized dominant peak locations $m_k/2^k$	$5 \leq k \leq 12$
renormalized_defects.png	$R(n) = Q(2n) - 2Q(n)$	$1 \leq n \leq 4096$
dyadically_normalized_R_logscale.png	dyadically normalized defect profiles	$2^4 \leq n \leq 2^{15}$
fourier_spectra_dyadic_profiles.png	$P_k(\ell) = \hat{R}_k(\ell) ^2$	$12 \leq k \leq 20$
even_odd_profiles_dyadic_blocks.png	parity-separated defect profiles	$k \in \{14, 16, 18, 20\}$

Table 3: Numerical ranges used for the main figures.

No windowing was used. The reported power fractions therefore refer to the raw dyadic block profiles after blockwise centering and normalization. As a robustness check, the Fourier analysis was repeated after subtracting separate even-site and odd-site means. It was also repeated on the parity subprofiles

$$R_k^{(a)}(m) = R(2^k + 2m + a), \quad a \in \{0, 1\}, \quad 0 \leq m < 2^{k-1}.$$

These parity checks are reported in Table 2 and Figure 10.

Verified identities and consistency checks

Several independent consistency checks were performed throughout the computations. On the computed ranges, we verified positivity of both recursive arguments $n - Q(n-1)$ and $n - Q(n-2)$, parity preservation $Q(n) \equiv 1 \pmod{2}$, and the first-difference constraint $Q(n+1) - Q(n) \in \{-2, 0, 2\}$.

We also checked the dyadic frequency identities on all complete verified dyadic blocks $B_k = [2^k, 2^{k+1})$. In particular, the computed frequencies satisfied

$$\{F(s) : 2^k \leq s < 2^{k+1}\} = \{3 + \nu_2(j) : 1 \leq j \leq 2^k\},$$

together with the block-mass identity

$$\sum_{s=2^k}^{2^{k+1}-1} F(s) = 4 \cdot 2^k - 1.$$

The cumulative counting function was checked against $C(2^k - 1) = 4(2^k - 1) - k$ and, more generally, against $C(m) = 4m - k - s_2(r+1)$, where $m = 2^k + r$ and $0 \leq r < 2^k$.

For the clock process $t_1(n) = n - Q(n - 1)$, we verified the reconstruction identity $Q(m) = m + 1 - t_1(m + 1)$. The defect sequence was computed both directly from $R(n) = Q(2n) - 2Q(n)$ and independently from the clock formula

$$R(n) = 2t_1(n + 1) - t_1(2n + 1) - 1.$$

The two computations agreed throughout the verified range. The main computations were also repeated using separately written scripts, with matching outputs on the overlapping ranges.

The numerical structures reported throughout the paper therefore arise directly from the recursive data rather than from smoothing, interpolation, or postprocessing artifacts.

C Alternative Initial Conditions and Exceptional Orbits

Although the present paper focuses primarily on the canonical initial condition $Q(1) = Q(2) = 1$, we briefly note that alternative initial seeds generate several qualitatively distinct dynamical behaviors. Most small initial conditions terminate rapidly because one of the recursive arguments becomes nonpositive, while a small number of seeds produce long-lived trajectories with stable large-scale growth behavior.

Proposition C.1 (An explicit rigid orbit). *Consider the perturbed Hofstadter recursion with initial values $Q(1) = 1$, $Q(2) = 1$, and $Q(3) = 2$. Then, for every $m \geq 2$,*

$$Q(2m) = 2m, \quad Q(2m + 1) = 2.$$

Proof. The claim follows by induction on m . For $m = 2$,

$$Q(4) = Q(4 - Q(3)) + Q(4 - Q(2)) + 1 = Q(2) + Q(3) + 1 = 4,$$

and

$$Q(5) = Q(5 - Q(4)) + Q(5 - Q(3)) - 1 = Q(1) + Q(3) - 1 = 2.$$

Assume now that $Q(2m) = 2m$ and $Q(2m + 1) = 2$ for some $m \geq 2$. Then

$$\begin{aligned} Q(2m + 2) &= Q((2m + 2) - Q(2m + 1)) + Q((2m + 2) - Q(2m)) + 1 \\ &= Q(2m) + Q(2) + 1 = 2m + 2, \end{aligned}$$

and similarly

$$\begin{aligned} Q(2m + 3) &= Q((2m + 3) - Q(2m + 2)) + Q((2m + 3) - Q(2m + 1)) - 1 \\ &= Q(1) + Q(2m + 1) - 1 = 2. \end{aligned}$$

This completes the induction. □

This orbit contrasts sharply with the irregular fluctuation dynamics observed for the canonical trajectory and demonstrates that the recursion also admits completely rigid self-sustaining solutions.

Initial seed	Behavior	Description
(1, 1, 1)	survives	canonical trajectory
(2, 1, 1)	survives	merges into canonical orbit
(3, 1, 1)	survives	merges into canonical orbit
(1, 3, 3)	survives	shift-equivalent to canonical orbit
(1, 1, 2)	survives	explicit rigid orbit
(1, 3, 1)	survives	long-lived irregular trajectory
(2, 1, 2)	survives	long-lived irregular trajectory
(2, 2, 1)	survives	long-lived irregular trajectory
(2, 3, 1)	survives	long-lived irregular trajectory
(3, 1, 2)	survives	long-lived irregular trajectory
most remaining seeds	terminate	rapid collapse

Table 4: Observed dynamical behavior for representative small initial triples. Most seeds terminate rapidly, while a small number produce long-lived or structured trajectories.

Small-seed classification

We numerically investigated all initial triples $(a, b, c) \in \{1, 2, 3\}^3$ under the perturbed recursion

$$Q(n) = Q(n - Q(n - 1)) + Q(n - Q(n - 2)) + (-1)^n.$$

The observed behaviors fall into several distinct dynamical classes. All listed surviving trajectories were verified numerically up to at least $N = 10^6$.

The existence of multiple long-lived trajectories suggests that the dyadic fluctuation mechanism is not restricted to a single exceptional initial condition. At the same time, the rarity of stable seeds indicates that the recursive dynamics remain highly sensitive to the initial configuration.

Several of the surviving trajectories appear numerically to share the same coarse asymptotic growth law $Q(n) \sim n/2$, while exhibiting substantially different local fluctuation geometry. This suggests that the global scaling behavior may be comparatively robust even though the finer recursive timing dynamics remain highly seed-dependent.

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